

Bayesian Optimisation of Compressive Properties in Bio-Derived Unsaturated Polyester Particulate Composites

7th July 2025

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Abstract

This paper will focus on particular group of thermosets, namely, polyester resins. These liquid resins generally cured to form the matrix that surrounds reinforcing materials within the structure of composite materials such as glass-reinforced plastic. This paper initially elucidates the method by which an entirely bio-based polyester resin was manufactured, from simple organic acids and alcohols. Subsequently, a description is provided of how Bayesian optimisation was applied to improve mechanical properties of interest by using a number of predictors to tune the sample compositions. In the multi-objective setting, it was found that these materials could have their yield strength and Young's modulus improved upon up to a putative global optimum of 165 MPa and 5,458 MPa respectively. In the single-objective setting, an experiment succeeded in developing the Young's modulus of these bio-based materials up to 5,012 MPa which equated to 69% the best modulus obtained from an identical optimisation of the equivalent 'conventional' styrene-based material.

1 Introduction

Thermoset composites underpin structural applications ranging from construction panels to transport components due to their high stiffness, dimensional stability and chemical resistance¹. However, unlike thermoplastics, crosslinked thermosets cannot be readily remelted or reshaped, and end-of-life options remain dominated by incineration and landfill². In 2019 alone, 353 Mt of plastic waste was generated globally, of which only 9 % was recycled, while thermoset-rich composite waste is increasingly directed toward energy recovery through combustion². As most commercial thermosets are derived from fossil feedstocks, their incineration reintroduces fossil carbon into the atmosphere, directly coupling materials performance to carbon-cycle perturbation³.

Since the vast majority of the virgin plastic manufactured annually is thermo-

plastic this group has naturally dominated plastic sustainability literature, with papers focused upon improvement of recycling methods⁴, mitigation of microplastic waste⁵, and biodegradability⁶. However, in part due to their relatively low industrial volume have received far less attention⁷. It is generally impractical to depolymerise crosslinked thermoset materials which has led to the widespread landfilling or incinerating of these materials at end-of-life⁸⁻¹⁰. Contemporary European legislation has prioritised incineration ahead of landfill for composite materials with thermosetting matrices, and incoming legislation (e.g. Directives '99/31/EC' and '2000/53/EC') is likely to increase the proportion incinerated¹¹. As the majority of thermoset plastics are derived from fossil-based sources of carbon, their combustion releases the geologically sequestered carbon into the atmosphere, resulting in a net addition of CO₂ into the

short-term carbon cycle¹². The present work addresses this challenge by initially identifying a suitable bio-based thermoset chemistry, whereafter it seeks to sample efficiently formulate these for maximal cured strength by a procedure involving Bayesian optimisation.

Bayesian Optimisation of Thermosets

In polymer science, Bayesian optimisation has been deployed in both purely computational domains (e.g. optimisation of thermoplastic repeat units for improved glass transition temperatures in a simulated parameter space¹³) as well as in applied practice (e.g. improving compositional homogeneity during co-polymerisation of styrene and methyl methacrylate¹⁴). Various thermosets have been practically optimised using Bayesian optimisation: Benzoxazine-based thermosets have been optimised with respect to six thermomechanical properties¹⁵, polycyanurates have had three thermal/mechanical/hygroscopic properties improved upon¹⁶, and epoxy thermosets based on amine curing agents have been optimised with respect to a trio of mechanical properties using a dual-driver method¹⁷. In the more specific domain of practical Bayesian optimisation of bio-based thermosets, literature has been dominated by epoxy-based chemistries; Albuquerque et al. successfully improved glass transition temperatures in bio-based epoxy thermosets derived from amino acid based resin systems¹⁸, whilst Rothenhäusler et al. explored this class of thermosets further by taking into account both cost and bio-based content¹⁹.

Bayesian optimisation has seen practical application in the development of thermoset composites with a paper by Schoenholz et al. considering unidirectional carbon fibre prepreg based on epoxy resin; here, both the layup and curing cycle variables were optimised to reduce the process-induced deformation in final composite parts²⁰. Despite this, relatively little literature exists concerning the practical application of Bayesian

optimisation in the development of properties in polyester-based thermosets, their bio-based analogues, or their derived composites. In applying Bayesian optimisation for the development of strength in simple composites derived from bio-based thermoset chemistries, this paper addresses this gap in the literature.

Bio-Based Thermoset Chemistry

Thermosets are formed from a wide range of different chemistries including (but not limited to): phenolic resins, unsaturated polyester resins, epoxy resins, and polyurethanes²¹. One prominent group are the unsaturated polyester resins, which are routinely used in the manufacture of composite components. These resins exhibit desirable properties including ease of handling in liquid form, rapid curing, and excellent dimensional stability^{22,23}. An archetypical unsaturated polyester resin is the mixture of poly(propylene glycol fumarate) copolymer (polymerised from propylene glycol and maleic anhydride) with a styrene reactive diluent²⁴. Unfortunately this would not be suitable as a substrate for bio-based composites, since all monomers associated with conventional unsaturated polyester resins are fossil-derived products. For example, propylene glycol is industrially manufactured from propylene oxide- a downstream chemical product of propane extracted from natural gas¹². Maleic anhydride is synthesised by catalytic oxidation of the chemical butane²⁵- this derived from fractional distillation of petroleum crude²⁶. And styrene is produced from the dehydrogenation of ethylbenzene- this being the product of a reaction between benzene and ethylene, both of which are fossil-fuel derived feedstocks^{12,27}.

Although literature describes numerous attempts at synthesising the aforementioned conventional monomers from bio-based feedstocks (e.g. maleic anhydride²⁸⁻³⁰ and styrene³¹) only research into propylene glycol³²⁻³⁵ has achieved apparent success in

scale-up and commercialisation, so we retained this monomer in the present work. Also incorporated into this work polyester synthesis was itaconic acid, a simple bio-based difunctional carboxylic acid manufactured by fungal fermentation of sugars by fungi such as *Aspergillus terreus*³⁶. Itaconic acid's dicarboxyl and monoalkene functionality allows it to directly substitute maleic acid during polycondensation to form an unsaturated polyester³⁷. Furthermore, itaconic acid can be synthesised into a variety of diesters (the family of diitaconates) which exhibit properties appropriate of a reactive diluent (i.e. radical initiated homopolymerisation and a low viscosity), allowing them to replace styrene in an eventual unsaturated polyester resin³⁸.

Itaconic acid's potential is clear in the literature, where subjects including the catalytic influence on polycondensation of itaconic acid within eventual unsaturated polyesters³⁹, synthesis of partially bio-based resins incorporating itaconic-acid-based unsaturated polyester but retaining styrene as the reactive diluent⁴⁰, as well as fully bio-based resins using itaconic-acid-based unsaturated polyester in tandem with diitaconates as reactive diluents^{41,42} have been disseminated. Furthermore, previous studies have considered partially bio-based resins incorporating itaconic-acid-based unsaturated polyesters together with the cross-linking compatible methacrylate class of reactive diluents^{43,44}. Although traditionally fossil-fuel derived, recent work has seen bio-based methyl methacrylate synthesised from itaconic acid, which would unlock fully bio-based resins based on methacrylate reactive diluents in future^{45,46}. Despite inferior homopolymerisation rates³⁸ and viscosity characteristics⁴³ relative to methacrylates, diitaconates were nonetheless focused upon in this research ahead of methacrylates on account of their relatively higher strengths as cured thermosets⁴³, safety as reactive diluents (lower toxicity and flammability)⁴⁷, and environmentally-benign nature as a non-

volatile organic compound^{43,48}.

A Thermoset Particulate Composite

In 1997, Kharas et al. prepared several samples of bioresorbable bone cement composites using both polyester-based thermosets in combination with particulate calcium sulphate, where the latter's content was found to effect the final thermosets' strength and modulus⁴⁹. The sensitivity of mechanical properties toward this variable inspired the present work's adoption of anhydrous calcium sulphate as a particulate reinforcing element during the preparation of simple particulate composites based on a fully bio-derived unsaturated polyester resin.

Counterparting this particulate reinforcing, the thermosetting matrix was based on a cross-linked unsaturated polyester system; these being reliant upon a combination of both condensation and addition polymer. The condensation polymer is typically an unsaturated linear polyester formed from the copolymerisation of diols and diacids, with at least one monomer having an unsaturated double bond in its structure which is retained after copolymerisation- this copolymer is referred to as an unsaturated polyester⁵⁰. Whilst the addition polymer is typically generated through a radical-addition homopolymerisation reaction, where a monomer which had been acting as an unsaturated polyester solvent reacts and crosslinks individual unsaturated polyester chains- these monomers are therefore referred to as reactive diluents⁵¹. The combination of unsaturated polyester and reactive diluent is known as an unsaturated polyester resin; a viscous liquid, which, on addition of radical-initiating chemicals, can rapidly form a rigid thermoset material (i.e. the composite's thermoset matrix). Previous work by Panic et al. found dimethyl itaconate to be the strongest contender of the diitaconates as a bio-based replacement for styrene in unsaturated polyester resins, given that resultant cured thermosets exhibited relatively higher crosslink densities

(73% the density of a styrene-based thermoset)⁴². This work therefore sought to replicate these unsaturated polyester resins, where the reactive diluent was pure dimethyl itaconate and the unsaturated polyester was poly(propylene itaconate).

Having both described the particular material considered in the present work and provided a thorough review of Bayesian optimisation’s utility in the sample-efficient formulation of thermoset materials, the paper will subsequently take the following form: Initially, a detailed description of experimental methods will be provided spanning feedstock synthesis, bio-derived unsaturated polyester resin cooking, and particulate composite preparation. Furthermore, this method section provides details of the various optimisation procedures implemented. Subsequently, a results and discussion section disseminates findings arising from both the single-objective and dual-objective routines undertaken, as well as cross-comparing the optimisation of a model conventional resin based on styrene. Finally, a conclusions section notes the more salient findings such as the successful single-objective Bayesian optimisation of these materials to strengths approaching those of styrene analogues.

2 Materials & Methods

Optimisation experiments carried out here involved both the practical methods associated with preparation and characterisation of thermoset composite samples as well as computational techniques developed to carry out Bayesian optimisation based on data obtained. Furthermore, synthetic routes are presented for producing any materials not readily available commercially. This section will first present practical methods including synthesis of itaconic acid polyester and diesters, thermoset composite preparation, and characterisation techniques employed. Subsequently, the computational method by which Bayesian optimisation was imple-

mented will be elucidated.

2.1 Materials

Propylene glycol (99%), itaconic acid (99%), anhydrous calcium sulphate (99%), and hydroquinone (99%) were obtained from “Thermo Scientific Chemicals”, a division of *Thermo Fisher Scientific Inc.* based in Waltham, Massachusetts, U.S. Toluene (99.8%), dried magnesium sulphate (99%), sodium chloride (99.5%), and dichloromethane (95%) were obtained from “Fisher Chemicals”, a division of *Thermo Fisher Scientific Inc.* Mixed xylenes (97%) were obtained from “Alfa Aesar” a division of *Thermo Fisher Scientific Inc.* Methanol (99.9%) was obtained from “Honeywell Riedel-de Haën”, a brand under *Honeywell International Inc.* based in Charlotte, North Carolina, U.S. P-toluenesulphonic acid monohydrate (98%) and chloroform-d (99.8%) were obtained from “Sigma-Aldrich”, a subsidiary of *Merck KGaA* based in Darmstadt, Germany. Dibenzoyl peroxide (20%) filled with calcium sulphate was obtained from *Nouryon Specialty Chemicals B.V.* based in Amsterdam, The Netherlands. Sodium bicarbonate (99%) was obtained from *Fluorochem Ltd* based in Glossop, Derbyshire, U.K. Phenolphthalein solution (1% w/w in ethanol) and potassium hydroxide (0.56% w/w in methanol) were supplied by *Scientific Laboratory Supplies Ltd* based in the East Riding of Yorkshire, U.K.

2.2 Diitaconate Synthesis & Characterisation

Dimethyl itaconate was synthesised by esterification of itaconic acid with methanol by the Fischer method which has previously been used successfully in the literature^{48,52,53}; this is presented in Figure 1 as a simplified reaction scheme. The method was adapted from the work of Trejbal et al. and involved the mixing of itaconic acid in an excess of methanol in a molar ratio of 1:3⁴⁸, whilst

p-toluenesulphonic acid was used as a heterogeneous strong acid catalyst and made up 4% w/w of the final reaction mixture. Detailed practical method by which this was achieved has been placed in the supporting information (Sec. 6.1), but may be shortly summarised: initially the reaction mixture is refluxed to produce the impure product, liquid-liquid extraction is then employed to draw out the pure product into a suitable organic solvent, and finally this solvent is removed under reduced pressure to yield the final product.

Several different characterisation techniques were used to monitor the reaction progress, identify chemicals, and ascertain the product purity, including thin layer chromatography (TLC), Fourier-transform infrared spectroscopy using attenuated total reflectance (ATR-FTIR), and ^1H and ^{13}C nuclear magnetic resonance spectroscopy (NMR). Preliminary investigations undertaken using TLC have been detailed in the supporting information (see Sec. 6.2) for the sake of brevity.

ATR-FTIR spectra were obtained using a Nicolet iS50 machine with the spectral range between $400\text{--}4000\text{ cm}^{-1}$ being investigated at a resolution of 0.09 cm^{-1} . Figure 2 presents the spectra obtained for the pure reactant (itaconic acid) and compares it with that of the product isolated after rotary evaporation. These data support the hypothesis that the reactant was successfully esterified to form the product dimethyl itaconate in a number of ways.

Firstly, the broad peak characteristic of O—H bond stretching commonly located between $2500\text{--}3200\text{ cm}^{-1}$ is clearly present for the itaconic acid but not for the product (Fig. 2A), suggesting the loss of O—H bonds and therewith the carboxyl groups associated with either the initial reactant (itaconic acid) or the intermediate product (monoethyl itaconate).

Secondly, strong peaks are observed around 1710 cm^{-1} (Fig. 2B & 2D) which is commonly indicative of C=O bond stretch-

ing (which occurs between $1640\text{--}1760\text{ cm}^{-1}$); this would be consistent with C=O bond retention which would occur during successful esterification. Furthermore, it may be noted that peaks associated with carboxylic C=O bonds occur between $1640\text{--}1730\text{ cm}^{-1}$ whilst ester C=O peaks may occur between $1690\text{--}1750\text{ cm}^{-1}$; the positive difference between the reactant C=O peak located at 1700 cm^{-1} (Fig. 2B) and the product's peak at 1720 cm^{-1} (Fig. 2D) would support successful esterification.

Thirdly, an intermediate-sized peak is observed both for the reactant (Fig. 2C) and the product (Fig. 2E) at 1630 and 1640 cm^{-1} respectively, which is suggestive of nonconjugated C=C bond stretching (occurring between $1620\text{--}1680\text{ cm}^{-1}$), which would be consistent with retention of the double bond during the proposed esterification. As an aside, clusters of peaks likely associated with C-H bond stretching were observed on both spectra (Fig. 2F & 2G) but inter-comparison was obfuscated by the strong and broad O—H stretching signal apparent in the reactant's spectrum.

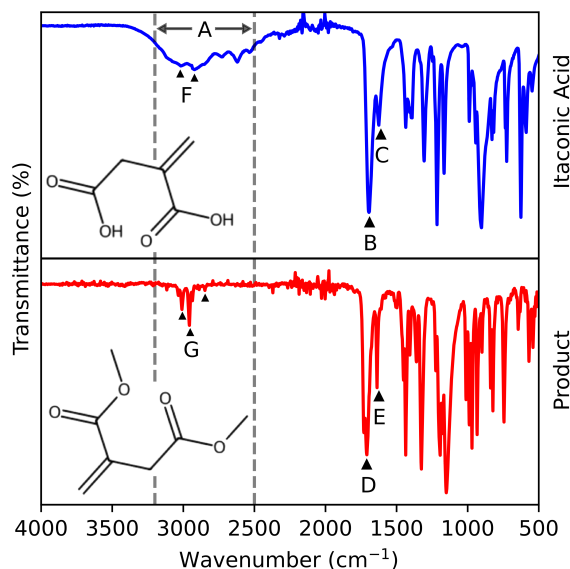


Figure 2: ATR-FTIR spectra of reactant and product. (A) O—H stretch at $2500\text{--}3200\text{ cm}^{-1}$. (B) C=O stretch at around 1700 cm^{-1} . (C) 1630 cm^{-1} . (D) C=O stretch at around 1720 cm^{-1} . (E) 1640 cm^{-1} . (F) 2950 & 3050 cm^{-1} . (G) 2875 , 2980 & 3035 cm^{-1} .

Both ^1H and ^{13}C NMR spectroscopy was carried out on a sample of product using

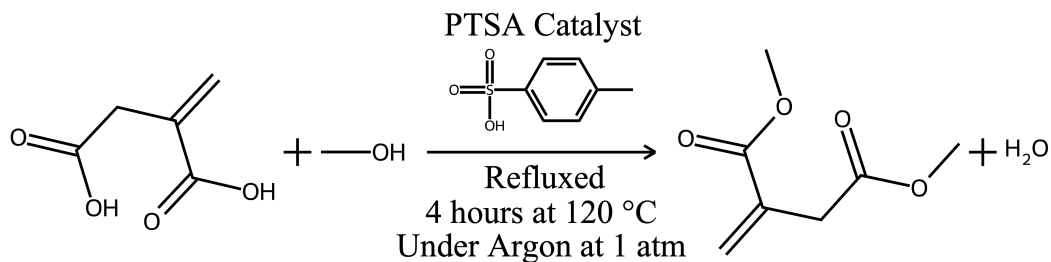


Figure 1: Simple reaction scheme for synthesis of dimethyl itaconate.

a Bruker Avance III HD 500 MHz Smart Probe spectrometer, with spectrograms produced at 500 MHz and 125 MHz respectively. Chemical shifts are considered as downfield shifts (δ in ppm) from the tetramethyl silane standard ($\delta = 0$ ppm) and calibrated using the shift exhibited by trace amounts of non-deuterated chloroform ($\delta = 7.26$ ppm singlet for ^1H and $\delta = 77$ ppm triplet for ^{13}C) within the sample solvent. The chloroform-d solvent also introduced a trace amount of water which exhibited a singlet peak ($\delta = 1.63$ ppm) in the ^1H spectrogram; these peaks were ignored when examining the spectra for dimethyl itaconate, as presented here:

Dimethyl itaconate: ^1H NMR (CDCl_3 , 500 MHz): δ 3.34 (dd, 2H, $J = 0.71, 0.15$ Hz), 3.69 (s, 3H), 3.76 (s, 3H), 5.71 (q, 1H, $J = 0.97$ Hz), 6.32 (dt, 1H, $J = 0.84, \text{n/a}$ Hz). $^{13}\text{C}\{^1\text{H}\}$ NMR (CDCl_3 , 125 MHz): δ 37.5, 52.1, 52.2, 128.6, 133.7, 166.6, 171.1.

A quantitative ^1H spectrogram showed water constituting a 5% w/w impurity, although the dimethyl itaconate sample purity is expected to exceed 95% given that the deuterated solvent will have introduced an unknown amount of water. This proved that the synthetic method devised was capable of producing dimethyl itaconate at a technical grade or above in terms of purity, allowing the subsequent experimental work to proceed.

2.3 Unsaturated Polyester Resin Synthesis & Characterisation

Unsaturated polyester was synthesised via the polyesterification of itaconic acid with propylene glycol to produce poly(propylene itaconate), as described by the simplified reaction scheme in Figure 3. This was achieved using the melt polycondensation method, since a precedent exists in the literature for its successful use in the synthesis of linear unsaturated polyesters derived from itaconic acid^{41–44}. The stoichiometric ratio between itaconic acid and propylene glycol was set at 1:1.1, on the basis that propylene glycol is lost during the heating process which reaches temperatures exceeding its boiling point⁵⁴.

Initially, the ‘azeotropic method’ of melt polycondensation was undertaken whereby a reaction vessel was charged with diols, diacids, and a water-entraining agent (5% w/w mixed xylenes) before being heated for an extended period (approximately 11 h). Subsequently, an inhibiting agent (0.06% w/w hydroquinone) was added to the molten polyester before combination with reactive diluent (30% w/w dimethyl itaconate) to produce the final unsaturated polyester resin. A more thorough treatment of this method may be obtained in the supporting information under Section 6.3. As a predictor variable which was to be used during optimisation, the reactive diluent content of resins produced initially was intentionally made lower than optimal ratios seen in lit-

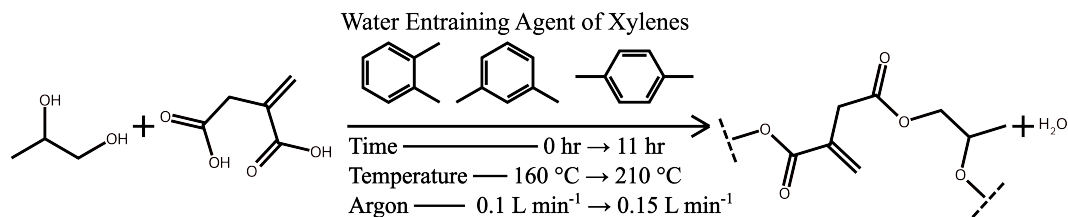


Figure 3: Simple reaction scheme for synthesis of poly(propylene itaconate).

erature^{24,54–56}, this allowed a range of resins with varying amounts of reactive diluent to be tested by simply mixing in more of these monomeric reagents to the initially produced resins.

Gel permeation chromatography (GPC) was used to separate species according to hydrodynamic volume in solution, this allowed us to obtain estimates for both number and weight average molecular weights as well as polydispersity of the synthesised unsaturated polyesters. Testing was undertaken using a Viscotec system (manufactured by Malvern Panalytical) consisting of both a GPCmax autosampling module as well as the TDA305 triple-detection module (providing a concentration detector, viscometer, and light scattering detector). The column set employed consisted of a guard column followed by two Agilent PLgel 5 μ m MIXED-D analytical columns, with these providing high-molecular weight resolution typically necessary for the study of resins. Consistent with previous GPC work on unsaturated polyesters we used tetrahydrofuran as the eluent, with this suitably solvating the unsaturated polyester samples obtained from the polyesterification reactor⁵⁷. Polyester samples were diluted to 5 mg mL⁻¹ and filtered through an Agilent 0.45 μ m polytetrafluoroethylene membrane before entry into the GPC columns.

Unfortunately, solvated polyester’s hydrodynamic viscosity depends upon various factors other than molecular weight (e.g. extent of branching or chemical composition) preventing a universal calibration of the GPC columns⁵⁸. Alternatively, a conventional generic calibration was per-

formed by running a series of polystyrene samples (Agilent EasiVial PS-H) of known molecular weights on the columns. Subsequently, a 3rd degree polynomial was fit and used convert a sample’s retention volume to a polystyrene-equivalent molecular weight. This relative calibration method has previously been shown to successfully approximate the number average molecular weight of polyesters, with values being in fairly good agreement with those obtained from either vapour pressure osmometry or end group analysis⁵⁸.

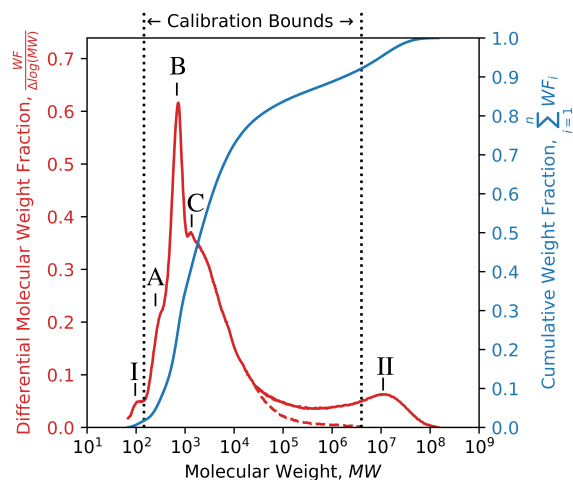


Figure 4: A graph depicting the distribution of mass across various polymer chain lengths for the polyester sample tested using GPC. The dashed red line describes a hypothetical deviation from the apparent distribution. I/II: Unphysical peaks from total permeation and exclusion. A/B/C: Identifiable peaks.

Figure 4 presents the results of the GPC characterisation of the batch of polyester synthesised, with both differential molecular weight fraction and cumulative weight fraction plotted against apparent molecular weight of chains. The apparent distribution is multi-modal with two peaks

occurring outside of the calibration limits (Fig. 4I/II) and three peaks identifiable within (Fig. 4A/B/C).

Peaks I and II are caused by total permeation and exclusion respectively, with both being reasonably expected to be unphysical as they occur outside the of the styrene-standard calibration bounds, rendering molecular weights estimated here uncertain extrapolations. Despite this, weight fraction associated with peak II remained physical, and it may be estimated that around 15% of the polyester mass in the sample exhibited a molecular weight in excess of the upper calibration bound of $3.9 \times 10^6 \text{ g mol}^{-1}$. It is posited that this mass fraction is associated with network polymer species formed by unwanted premature crosslinking between branched polyester structures. Consequently, calculation of number M_n and weight M_w average molecular weights directly from the apparent distribution proffers values (952 & $1.6 \times 10^6 \text{ g mol}^{-1}$) far in excess of that which might be expected for either conventional maleic-based (709 & $3,382$) or similar itaconate-based polyesters (600 & $3,657 \text{ g mol}^{-1}$)³⁷. By ignoring the mass fraction which had undergone premature crosslinking (see Sec. 6.4) a hypothetically uncrosslinked distribution was plotted (red dashed line, Fig. 4). Not only did this distribution display a 15% reduction in its summed weight fraction relative to the apparent distribution, it also presented M_n and M_w values (606 & $3,006 \text{ g mol}^{-1}$) similar to those detailed in the literature³⁷.

Table 1: Summary of molecular weights and polydispersity values for both the apparently crosslinked, hypothetically uncrosslinked, and deconvoluted (A/B/C) distributions obtained from the GPC of the polyester resin.

By discerning the existence of three separate peak structures (Fig. 4A/B/C), mathematical deconvolution was employed (see Sec. 6.4) to split apart the apparent distribution into three log normal distributions allowing the estimation of M_n , M_w , weight fraction, and polydispersity D as summarised in Table 1. Distribution C represented the largest weight fraction of the modelled distributions, and exhibited polydispersity ($D = 5.62$) somewhat above those commonly reported for traditional unsaturated polyester resins (e.g. $D = 2$ ⁵⁹). Such dispersity values were likely the result of an extended reaction time (the outcome of eschewing catalysts) coupled with the lack of radical inhibiting agent, which would have facilitated a high degree of branching. Contrastingly, the modelled distributions A/B presented low dispersity values. These two peaks were consistent with a proportion of the linear polyester species undergoing intramolecular cyclisation.

Cyclised species present as a separate peak of lower molecular weight as their more compact conformation has a lower hydrodynamic volume relative to a linear species of the same molecular mass. Traditionally, polyesters were known to either undergo ‘ring closing’ or ‘back biting’ cyclisation, where respectively an end carboxyl group would react with either an end hydroxyl group or a mid-molecule ester group (see Figure 5). Literature presents this occurrence in GPC analyses for a variety of unsaturated polyester resins based on maleic and phthalic anhydride^{58,60,61}. It is also the case that intramolecular cyclisation may take place by an ‘Ordelt’ reaction, whereby an end hydroxyl group may react with a mid-molecule unsaturated group to form an ether linkage (see Figure 5). Species produced by the Ordelt mechanism differ from the pre-

Distribution	M_n	M_w	D
Crosslinked	952	1,559,183	1,638.00
Uncrosslinked	606	3006	4.96
A	165	168	1.02
B	837	872	1.04
C	1,727	9,711	5.62

vious two in that they retain a ‘molecular tail’ which ensures a relatively higher hydrodynamic volume given the same molecular weight species (see Sec. 6.4). Given hydrodynamic volume’s impact upon elution time and therewith estimated molecular weights presented in Figure 4, we believe that peaks B and A would be associated with those cyclised species produce via the Ordelt and traditional mechanisms respectively. Interestingly, when weight fractions modelled by the log normal distributions are considered (and crosslinked material therefore disregarded), it becomes apparent that 94 % of the sample mass was linear or branched (peak C), 5% was Ordelt cyclised (peak B), and 1% was traditionally cyclised (peak A). Narrow polydispersities associated with the two cyclised peaks may be due to narrow windows of stability associated with these conformations. For example, smaller cycles are known to have their stability negatively impacted by relatively higher cycle stress⁶².

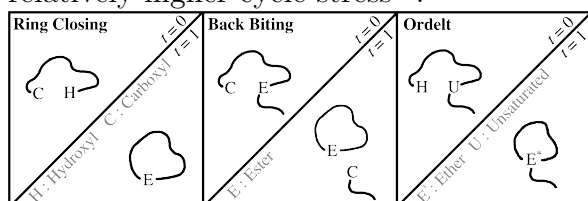


Figure 5: An illustration of the three mechanisms by which intramolecular cyclisation may be expected to occur in unsaturated polyester.

2.4 Composite Preparation

Figure 6 presents an overview of the method by which cured thermoset composite samples of various compositions were prepared. Given the danger of self-accelerating decomposition reactions and associated thermal runaway, we obtained a safely filled base initiating agent comprised of both dibenzoyl peroxide and calcium sulphate (Fig. 6A); this arrived pre-mixed in a set ratio of 2:8 between the dibenzoyl peroxide ($1 - c_2 = 0.2$ w/w) and the calcium sulphate ($c_2 = 0.8$ w/w). Base unsaturated polyester resin was mixed (Fig. 6B) in a set ratio of 3:7 between unsaturated polyester resin ($c_1 = 0.7$ w/w) and dimethyl itaconate ($1 - c_1 = 0.3$ w/w)

as described previously. This elected ratio represents the lowest amount of reactive diluent possible before increasing viscosity precluded the possibility of further mixing. This increased the size of the possible parameter space to be interrogated by increasing the range of modified unsaturated polyester resin reactive diluent concentrations that could be mixed to by means of dilution.

Modified unsaturated polyester resin was mixed (Fig. 6C) in a varying ratio of $x_1 : (1 - x_1)$ between the base unsaturated polyester resin (x_1 w/w) and the dimethyl itaconate ($1 - x_1$ w/w). By varying the x_1 parameter over the range ($[0.45, 1.00]$) we could produce any of the desired mixing ratios, fully describing the first predictor variable to be optimised (x_1 , unsaturated polyester content modifier). Modified initiating agent was mixed (Fig. 6D) in a varying ratio of $x_2 : (1 - x_2)$ between the base initiating agent (x_2) and calcium sulphate ($1 - x_2$). The parameter x_2 was varied over the range ($[0.4, 1.00]$) hence describing the second predictor variable to be optimised (x_2 , dibenzoyl peroxide content modifier). Initiated unsaturated polyester resin was mixed (Fig. 6E) in a varying ratio of $x_3 : (1 - x_3)$ between modified unsaturated polyester resin (x_3) and modified initiating agent ($1 - x_3$); x_3 was provided a range ($[0.5, 0.95]$) which therewith described the final predictor to be optimised (x_3 , initiating agent content modifier).

Initiated unsaturated polyester resin samples were thermochemically cured into rigid thermoset plastic samples by exposing them to elevated temperatures in a drying oven (Fig. 6F); the curing profile was fixed for all samples: 3 hours at 80°C followed by 1 h at 120°C. All cured thermoset samples were machined to the same dimensions (Fig. 6G) in preparation for characterisation; this was achieved by a grinding process.

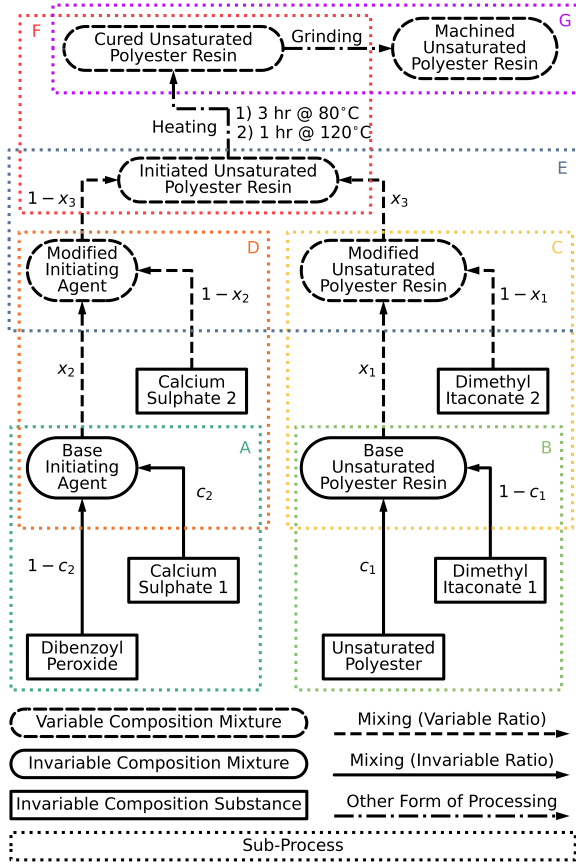


Figure 6: *Mixing and processing schedule for the manufacture of thermoset composite samples. (A) Manufacturer’s mixing for base initiator. (B) In-house mixing for base resin. (C) Mixing for modified resin. (D) Mixing for modified initiator. (E) Mixing for initiated sample. (F) Thermochemical curing. (G) Machining.*

2.5 Composite Characterisation

In order to quantify the mechanical performance of thermoset polymers created using Bayesian optimisation, their elastic modulus and yield strength were characterised using compressive testing. Compressive testing was chosen based on both the simplicity with which a curing regime could be devised for the manufacture of samples of the requisite dimensions and the low mass of material required for carrying out tests. An American Society for Testing and Materials standard was employed (“ASTM D695-23”) in the devising of method, with a cylindrical test sample diameter of 12.7 mm height 12.7 mm, and a compression speed of 1.3 mm min^{-1} adopted. To mitigate against concerns of

Euler buckling taking place in samples of the proposed dimensions, we calculated the critical load based on Euler’s critical load equation (see supporting information) and some estimated properties for conventional polyester thermosets⁶³. What follows is an abridged review of the method undertaken in the automated analysis of datasets derived from mechanical testing, with more detail provided in the supporting information.

Equipment Compressive testing was carried out using a “Tinius Olsen 50ST” universal test machine (50 kN load capacity) fitted with compression platens. Standard software (‘Horizon’) purveyed by the machine’s manufacturer was used in the control of the machine, retrieval of data from the load cells, and export of formatted data for further analysis. Exported data consisted of time ($\pm 0.16 \text{ } \mu\text{s}$), force ($\pm 1 \text{ mN}$), and platen displacement ($\pm 1 \text{ nm}$).

Correction of Raw Data Finite compliance of the testing machine introduced an additional displacement contribution into raw exported datasets. To correct for this, a blank compression test was performed with no sample positioned between the platens, generating a force-displacement curve corresponding to the inherent compliance of the machine. This curve was subsequently used as a correctional dataset. Figure 7 compares the correctional dataset with the representative raw and corrected compression datasets. Point ‘A’ on the raw dataset is transformed to point ‘B’ on the corrected dataset by subtracting the corresponding machine displacement contribution d_{cor} , from the measured displacement d_{raw} .

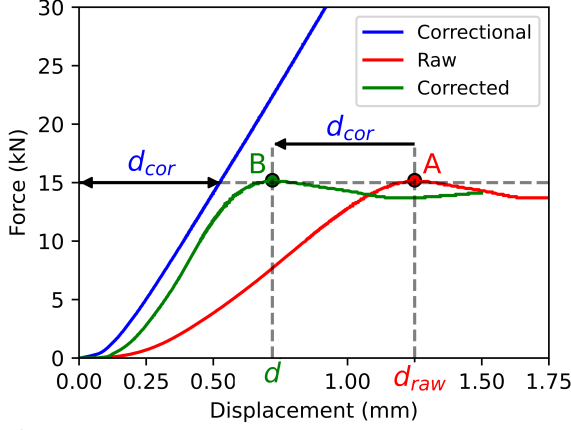


Figure 7: Strain correction of an archetypical raw dataset; the legend notes the various datasets considered.

Processing of Corrected Data Load values from a sample’s corrected dataset were converted into engineering stress values by dividing by the initial surface area. Each cylindrical sample’s diameter was measured using a “Magnusson” digital caliper (accuracy ± 0.02 mm) from three different positions (ϕ_1, ϕ_2, ϕ_3 in mm), each offset by roughly $+45^\circ$ around the circumference, and the mean of these values (ϕ_{avg} in mm) calculated. Subsequently, the average top-face surface area (A_{avg}) was obtained from the average diameter ϕ_{avg} , allowing the engineering stress (σ in MPa) to be calculated (Equation 1) by considering the force applied (F in N) and the initial top surface area, A_{init} ⁶⁴.

$$\sigma_{\text{eng}} = \frac{F}{A_{\text{init}}} \quad (1)$$

Displacement values from a sample’s corrected dataset were converted into engineering strain by taking the difference between the initial length of the samples and the length of the sample during compression. The initial length (l_{init} in mm) of sample was calculated by taking the mean of three length measurements (l_1, l_2, l_3 in mm) (these taken in the same fashion as for diameter). Following this, by considering the length of a sample at any given point in time during a compressive test ($l = l_{\text{init}} - d$), the engineering strain was calculated by finding the absolute change in sample length (Δl_{init} in

mm) and dividing this by the initial length of the sample (l_{init}) as shown in Equation 2.

$$\epsilon_{\text{eng}} = \frac{\Delta l_{\text{init}}}{l_{\text{init}}} = \frac{l - l_{\text{init}}}{l_{\text{init}}} \quad (2)$$

Stress-Strain Smoothing To produce a smoothed dataset (from which Young’s modulus and yield strength could be deduced) a combination of sample-rate reduction and filtering were applied to downsample the stress-strain dataset. Downsampling was achieved using a windowed averaging technique, whereby the unsmoothed dataset was split into a number of windows along the strain-axis by electing a suitable number of points per window. Figure 8 demonstrates the efficacy of this technique in removing noise from the unsmoothed stress-strain dataset.

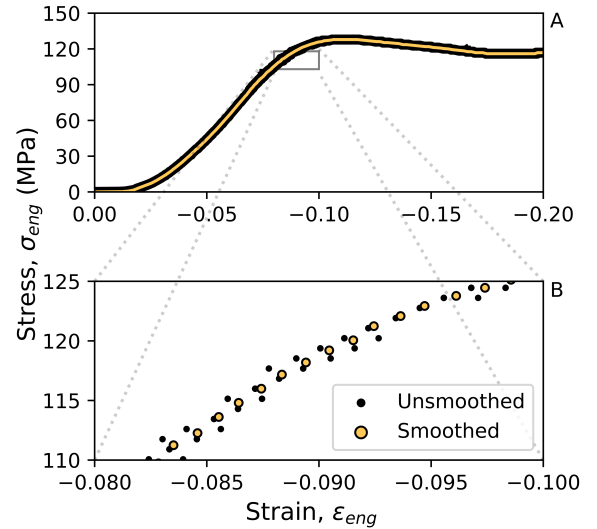


Figure 8: Multi-scale depiction of the stress-strain dataset before and after downscaling. (A) Overview line plot. (B) Zoomed scatter plot.

Derivative Analysis The proportional limit is defined as the stress at which a material ceases to obey Hooke’s law (i.e. stress no longer increases proportionally with strain). The gradient of the preceding elastic region of deformation is generally defined as the Young’s modulus. However, given that the stress-strain curves did not consistently exhibit a linear region, the limit of proportionality was equated to the point at which the stress-strain curve began to level off (i.e. the first point of inflection). To find this

inflection point, the first derivative of stress with respect to strain (i.e. tangent modulus E_t in MPa) was calculated from the smoothed stress-strain dataset by taking into account the instantaneous change in both stress ($\Delta\sigma_{\text{eng}}$) and strain ($\Delta\epsilon_{\text{eng}}$), and given by Equation 3.

$$E_t = \frac{\Delta\sigma_{\text{eng}}}{\Delta\epsilon_{\text{eng}}} = \frac{\sigma_{\text{eng}}^{j+1} - \sigma_{\text{eng}}^j}{\epsilon_{\text{eng}}^{j+1} - \epsilon_{\text{eng}}^j} \quad (3)$$

Figure 9 shows how smoothing of the stress-strain dataset facilitated the analysis of the resultant tangent modulus versus strain dataset. Step-like features are apparent in the tangent-modulus curve obtained from the unsmoothed stress-strain data. These may be consistent with either distretisation or internal processing of the exported displacement signals and subsequently amplified by the process of numerical differentiation. To avoid undue physical significance being assigned to these artefacts, proportional-limit detection was performed upon a smoothed stress-strain dataset.

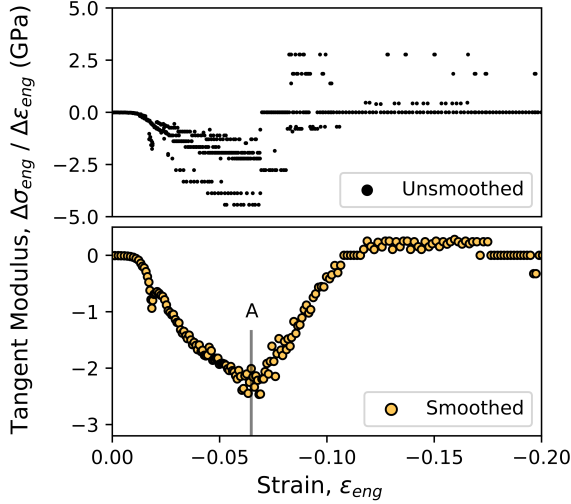


Figure 9: A comparison of the tangent modulus vs strain curves obtained from both the raw and the smoothed counterpart stress-strain datasets. (A) Strain at first inflection point.

Detection of the first inflection point was automated using an algorithm that iterated chronologically through the tangent modulus versus strain dataset from low to high strains. At each point, the algorithm calculates an average gradient exhibited by an equal number of points both before and after

the current strain value (the ‘point horizon’). Using a point horizon of ten, the first inflection point was defined as the first point at which the summed forward and backward gradients fell below zero. This point was taken as the proportional limit, as shown in Figure 9A.

Objective Retrieval By re-considering the smoothed stress-strain dataset, the point with the closest proximity to the proportional limit’s determined strain value was equated to the first inflection point: the proportional limit (Fig. 10A).

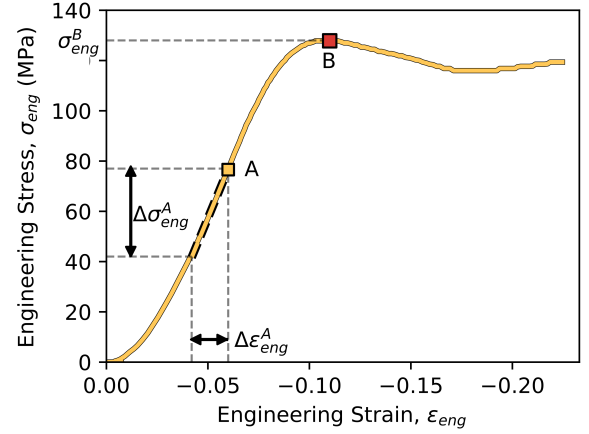


Figure 10: Automatic detection of Young’s modulus and yield point from an archetypal stress-strain dataset. (A) First inflection point. (B) Second inflection point.

As shown in Figure 10, a 1st degree polynomial was fit to the ten points preceding the proportional limit; allowing approximation of Young’s modulus (E in MPa) from the change in stress (Fig. 10: $\Delta\sigma_{\text{eng}}^A$) and strain (Fig10: $\Delta\epsilon_{\text{eng}}^A$) in accordance with Equation 4⁶⁵. Since the data were obtained in compression, the fitted gradients were negative; reported modulus values are given as absolute magnitudes.

$$E = \left| \frac{\sigma_{\text{eng}}}{\epsilon_{\text{eng}}} \right| = \left| \frac{\Delta\sigma_{\text{eng}}^A}{\Delta\epsilon_{\text{eng}}^A} \right| \quad (4)$$

The stress associated with the onset of non-linear deformation σ_y was used as an operational measure of compressive yield strength. In this work, this was identified as the engineering stress at the second inflection point of the stress-strain curve, shown

as σ_{eng}^B in Figure 10.

$$\sigma_y = \sigma_{\text{eng}}^B \quad (5)$$

For the stress-strain curves obtained in this study, this point coincided with the maximum engineering stress and could therefore be identified reproducibly across samples.

2.6 Optimisation

As described above, three predictors controlled the composition of samples: the unsaturated polyester content ratio (x_1), the dibenzoyl peroxide content modifier (x_2) and the initiating agent content modifier (x_3). These three predictors and their ranges formed a d -dimensional ($d = 3$) parameter space ($\mathcal{X}$) which is a subset of the entire real coordinate space ($\mathcal{X} \subseteq \mathbb{R}^d$). This parameter space was explored by the sequential optimisation techniques, with the objective variables including Young’s modulus (E) and yield strength (σ_y).

Three different optimisation procedures were carried out from the same initial prior dataset: two were single-objective Bayesian routines with differing objectives, whilst the third was a purely exploratory method which was objective-indifferent. A fourth multi-objective Bayesian optimisation routine incorporating both the objectives previously considered was carried out using a combination of the original prior dataset and all data generated by the three sequential optimisations.

Single-Objective Experiments Optimisation problems were based on black-box functions f which took any point within the parameter space $\mathcal{X}$ and returned a real-valued target $\mathbb{R}$; this being a mapping between the two respective spaces ($f : \mathcal{X} \rightarrow \mathbb{R}$). Equation 6 describes optimisation as the maximisation of the black-box function by tuning the predictor variable values ($\mathbf{x} = [x_1, x_2, x_3]$) such that the global optimum set of predictors ($\mathbf{x}^* = [x_1^*, x_2^*, x_3^*]$) is sought.

$$\mathbf{x}^* = \arg \max_{\mathbf{x} \in \mathcal{X}} f([x_1, x_2, x_3]) \quad (6)$$

Equation 7 describes the two black-box functions which were separately maximised during the course of the single-objective optimisations; the former f_E and latter f_σ considering Young’s modulus E and yield strength σ_y as target variables respectively.

$$\begin{aligned} E &= f_E(\mathbf{x}) \\ \sigma_y &= f_\sigma(\mathbf{x}) \end{aligned} \quad (7)$$

Eight vectors of predictor variables, each corresponding to a sample (i), were representatively drawn by Sobol sampling from across the parameter space $\mathcal{X}$ forming an initial matrix of predictors ($\mathbf{X}$); the matrix’s form is presented in Equation 8.

$$\begin{aligned} \mathbf{X} &= [\mathbf{x}^{i=1}, \mathbf{x}^{i=2}, \dots, \mathbf{x}^{i=8}] \\ &= [[x_1^{i=1}, x_2^{i=1}, x_3^{i=1}], [x_1^{i=2}, \dots], \dots] \end{aligned} \quad (8)$$

Evaluation of all the vectors of the predictor matrix ($f_E(\mathbf{X})$ or $f_\sigma(\mathbf{X})$) generated a vector of targets ($\mathbf{E}$ or $\boldsymbol{\sigma}_y$). Combination of the initial predictor matrix $\mathbf{X}$ and the corresponding targets $f(\mathbf{X})$ produced an initial dataset ($\mathcal{D} = [\mathbf{X}, f(\mathbf{X})]$) with which sequential optimisation experiments were begun. During the course of a sequential optimisation experiment, 16 further samples were drawn from across the parameter space $\mathcal{X}$, with one sample taken per iteration; each sample’s predictor vector $\mathbf{x}$ and function evaluation $f(\mathbf{x})$ was appended to the growing dataset $\mathcal{D}$ associated with that experiment. At each iteration, the Bayesian experiments strove to balance exploration and exploitation during selection of sample composition within the parameter space, whilst the benchmark deterministic experiment solely focussed upon exploration.

Bayesian optimisation involved an iterative process of developing a surrogate model approximating the latent objective function based on the available dataset $\mathcal{D}$, using an

acquisition function to evaluate candidate suitability and selection of a new sample $\mathbf{x}^i$, and finally evaluation of a sample followed by incorporation of the resultant target scalar σ_y^i and predictor vector $\mathbf{x}^i$ into the growing dataset $\mathcal{D}$.

On updating the dataset $\mathcal{D}$, Gaussian process regression was used to develop a Gaussian process regression model $\mathcal{GP}$ which can be considered a distribution over possible functions ($f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), K(\mathbf{x}, \mathbf{x}'))$) fully described by both its mean (m) and covariance (K) functions. Computationally, the Gaussian process was modelled by a high-dimensional multivariate Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ described by its mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$; each successive iteration saw the prior process conditioned on new data producing an updated posterior process.

To encode our prior beliefs in process smoothness, we employed the Matérn family of covariance functions; specifically, one with a hyperparameter $\nu = 5/2$ which yields Equation 9 based on Euclidean distance (d). The $\nu = 5/2$ case balanced the need for smoothness in sampled paths with the drawbacks of using higher values of ν . As $\nu \rightarrow \infty$, the sampled paths become infinitely differentiable, making them unrealistically smooth for practical application⁶⁶ and potentially interfering with successful calculation of the likelihood⁶⁷.

$$K_{M^{5/2}}(x, x') = (1 + \sqrt{5}d + \frac{5}{3}d^2)\exp(-\sqrt{5}d) \quad (9)$$

where $d = |x - x'|$

Each iteration saw the use of a one-step lookahead sampling policy utilising a simple reward utility function (Equation 10) which takes the current dataset $\mathcal{D}$ and of all samples $\mathbf{X}$ returns the sample $\mathbf{x}$ exhibiting the highest target (ϕ ; in practice either E or σ_y) according to the posterior process's mean function $\max m(\mathbf{X})$.

$$u(\mathcal{D}) = \max m(\mathbf{X}) \quad (10)$$

Utility quantification of a putative sample $\mathbf{x}_p$ was calculated by the expected improvement acquisition function α_{EI} (Equation 11) which calculates the marginal gain

in utility ($u(\mathcal{D}') - u(\mathcal{D})$) between the incumbent best target value in the current dataset (ϕ^*) and that of a new dataset containing the putative $\mathbf{x}_p$ value ($u(\mathcal{D}')$). The second line of Equation 11 clarifies how the calculation is one for the expected marginal gain, whilst the third presents reformulation for the noiseless case; this equation exemplifies how putative points are rewarded based on both scale and probability of improvement effectively trading off exploitation and exploration. Here, we applied a more sophisticated logarithmic noise-aware implementation (“qLogNoisyExpectedImprovement” or “qLNEI”) which has recently seen successful use in high-dimensional computational domains⁶⁸.

$$\begin{aligned} \alpha_{EI}(\mathbf{x}_p, \mathcal{D}) &= u(\mathcal{D}') - u(\mathcal{D}) \\ &= \mathbb{E}[\max m_{\mathcal{D}'}(\mathbf{X}')] - \max m_{\mathcal{D}}(\mathbf{X}) \\ &= \int_{\phi^*}^{\infty} \phi \mathcal{N}(\mu, \sigma^2) d\phi - \phi^* \int_{\phi^*}^{\infty} \mathcal{N}(\mu, \sigma^2) d\phi \end{aligned} \quad (11)$$

Trivial optimisation of the acquisition function (Equation 12) was achieved by generating a large set of Sobol sampled points across the parameter space $\mathbf{X}_p$, after which cheap evaluation facilitated identification of the candidate sample with the maximal utility score $\mathbf{x}^+$.

$$\mathbf{x}^+ = \arg \max_{\mathbf{X}_p \in \mathcal{X}} \alpha(\mathbf{X}_p) \quad (12)$$

The objective-indifferent space-filling benchmark sampling exercise was based on a sequential Monte Carlo strategy by the name of mc-intersite-proj-th (“MIPT”)⁶⁹. Utility was quantified by the conditional acquisition function displayed in Equation 13, which took both a putative point $\mathbf{x}_p$ and the current dataset's previous sample coordinates $\mathbf{X}$ and returns either zero or the Euclidean distance to the nearest previously sampled point k depending on whether projected distance is below or in excess of the d_m value respectively. The d_m is dependent upon the number of previous samples n and the arbitrary constant s which has been found to be optimally set to 0.5 for a reasonable trade-off between intersite distance and projected distance.

$$\begin{aligned} \alpha_{mipt}(\mathbf{x}_p, \mathbf{X}) &= \begin{cases} 0 & \text{if } \min_{\forall \mathbf{x}_i \in \mathbf{X}} \|\mathbf{x}_i - \mathbf{x}_p\|_{-\infty} < d_m \\ k, & \text{if } \min_{\forall \mathbf{x}_i \in \mathbf{X}} \|\mathbf{x}_i - \mathbf{x}_p\|_{-\infty} \geq d_m \end{cases} \quad (13) \\ \text{where } d_m &= \frac{2s}{n} \text{ and } k = \min_{\forall \mathbf{x}_i \in \mathbf{X}} \|\mathbf{x}_i - \mathbf{x}_p\|_2 \end{aligned}$$

Multi-Objective Experiment Given the desirability of both Young’s modulus and yield strength in a thermoset composite, a dual-objective Bayesian optimisation experiment was attempted, with the *a posteriori* goal of sample-efficiently identifying the Pareto front ($\mathcal{P}$)⁷⁰. The Pareto front consists of all samples which remain unsurpassed (by any other sample evaluated) in terms of at least one of their target variables (E or σ_y). This Pareto front allows a rational agent to make subjective yet informed decisions about material composition in light of expected trade-offs in end properties. Equation 14 frames the multi-objective optimisation problem in terms of a dual-objective function $\mathbf{f}$ (comprised of the single-objective functions f_E and f_σ) which returns a vector $\mathbf{y}$ of the two targets (E and σ_y).

$$\begin{aligned} \mathbf{y} &= \mathbf{f}(\mathbf{x}) \\ \{E, \sigma_y\} &= \{f_E(\mathbf{x}), f_\sigma(\mathbf{x})\} \end{aligned} \quad (14)$$

An observed Pareto front may be obtained directly from a given dataset $\mathcal{D}$ using the function $f_{\mathcal{P}}$ (Eq. 15).

$$\begin{aligned} \mathcal{P} &= f_{\mathcal{P}}(\mathcal{D}) \\ &= \{\mathbf{y} | \mathbf{y} \in \mathbf{Y}, \nexists \mathbf{y}' \in \mathbf{Y} \text{ s.t. } \mathbf{y}' \succ \mathbf{y}\} \end{aligned} \quad (15)$$

Alternatively, where a multi-objective function $\mathbf{f}$ is considered alongside a matrix of predictor variables $\mathbf{X}$, the function $f_{\mathcal{P}}^*$ may be used to produce a Pareto front (Eq. 16).

$$\begin{aligned} \mathcal{P} &= f_{\mathcal{P}}^*(\mathbf{X}; \mathbf{f}) \\ &= \{\mathbf{f}(\mathbf{x}) | \mathbf{x} \in \mathbf{X}, \mathbf{f}(\mathbf{x}) \succ \mathbf{f}(\mathbf{x}') \forall \mathbf{x}' \in \mathbf{X}\} \end{aligned} \quad (16)$$

By retrieving all the raw mechanical test data associated with the samples evaluated in the three sequential experiments and their common prior, a 56-sample dataset $\mathcal{D}$ was generated comprised of a matrix of sampled predictors $\mathbf{X}$ and a matrix of their respective targets $\mathbf{Y}$ ($\mathbf{E}$ and σ_y). Having conditioned two separate Gaussian processes on $\mathbf{E}$ and σ_y we obtained two separate posterior processes ($\mathcal{GP}_E$ and $\mathcal{GP}_\sigma$); these provided initial prior processes for all subsequent sequential optimisation. The pair of resulting processes were respectively described mathematically

by their mean (m_E, m_σ) and covariance functions (K_E, K_σ) and computationally by their multivariate normal analogues ($\boldsymbol{\mu}_E, \boldsymbol{\mu}_\sigma$ and $\boldsymbol{\Sigma}_E, \boldsymbol{\Sigma}_\sigma$). Equation 17 shows how a dual-objective surrogate function $\tilde{\mathbf{f}}$ may be sampled from the pair of Gaussian processes ($\mathcal{GP}_E$ and $\mathcal{GP}_\sigma$) previously conditioned.

$$\begin{aligned} \tilde{\mathbf{f}} &\sim \{\mathcal{GP}_E, \mathcal{GP}_\sigma\} \\ \{\tilde{f}_E, \tilde{f}_\sigma\} &\sim \{\mathcal{GP}(m_E, K_E), \mathcal{GP}(m_\sigma, K_\sigma)\} \end{aligned} \quad (17)$$

The utility of a given Pareto front was evaluated using the hypervolume indicator utility function u_{hv} presented in Equation 18, which is equal to the d -dimensional Lebesgue measure λ_d of the space dominated by a Pareto front $\mathcal{P}$ and bounded from below by reference point r (r and v describe vertices).

$$u_{\text{hv}}(\mathcal{P}; r) = \lambda_d\left(\bigcup_{v \in \mathcal{P}} [r, v]\right) \quad (18)$$

Equation 19 shows how the utility function u_{hv} is used in the subsequent calculation of the one-step marginal gain known as the “hypervolume improvement” between the extant $\mathcal{P}$ and the updated Pareto front $\mathcal{P}'$ (which includes the putative point $\mathbf{x}_p$).

$$f_{\text{hvi}}(\mathcal{P}, \mathcal{P}'; r) = u_{\text{hv}}(\mathcal{P}'; r) - u_{\text{hv}}(\mathcal{P}; r) \quad (19)$$

Here, we use a common multi-objective acquisition function called expected hypervolume improvement, described by Equation 20, which takes a putative point $\mathbf{x}_p$ and the current dataset $\mathcal{D}$ and returns the instantaneous improvement metric. Expectation manifests itself during the speculative update of the dataset $\mathcal{D}$ to $\mathcal{D}'$, where the two targets are drawn from their respective marginal Gaussian distributions (of their respectively updated posterior Gaussian processes) located at the putative point $\mathbf{x}_p$ whose expectation ($\mathbb{E}(\mathbf{x}_p)$) is the mean of the marginal at that location ($m_E(\mathbf{x}_p)$ or $m_\sigma(\mathbf{x}_p)$).

$$\begin{aligned} \alpha_{\text{ehvi}}(\mathbf{x}_p; \mathcal{D}) &= f_{\text{hvi}}(f_{\mathcal{P}}(\mathcal{D}'), f_{\mathcal{P}}(\mathcal{D}); r) \\ \text{where } \mathcal{D}' &= \mathcal{D} \cup \{\mathbf{x}_p, m_E(\mathbf{x}_p), m_\sigma(\mathbf{x}_p)\} \end{aligned} \quad (20)$$

This work adopted Daulton’s reformulation of the classic acquisition function (Eq. 20) which takes into account the

noisy nature of objective function evaluations⁷¹. Although the true formulation contains an integral which is analytically intractable, it may be approximated using Monte Carlo integration. Equation 21 describes the integral-approximated reformulation of the acquisition function $\hat{\alpha}$ which takes a putative point $\mathbf{x}_p$ and the dataset $\mathcal{D}$ and returns an instantaneous improvement value. This noisy expected hypervolume improvement acquisition function iterates from $1, \dots, N$ using the t counter, with each iteration drawing a random dual-objective function $\tilde{\mathbf{f}}_t$ from the posterior gaussian process (see Eq. 17). This function $\tilde{\mathbf{f}}_t$ is used to speculatively produce Pareto fronts ($\mathcal{P}$ and $\mathcal{P}'$) for both an original dataset $\mathbf{X}$ and the updated dataset $\mathbf{X} \cup \mathbf{x}_p$ respectively, before the hypervolume improvement is gauged using f_{hvi} and the many trials N averaged.

$$\hat{\alpha}_{\text{nehvi}}(\mathbf{x}_p; \mathbf{X}) = \frac{1}{N} \sum_{t=1}^N f_{\text{hvi}}(f_{\mathcal{P}}^*(\mathbf{X} \cup \mathbf{x}_p; \tilde{\mathbf{f}}_t), f_{\mathcal{P}'}^*(\mathbf{X}; \tilde{\mathbf{f}}_t); r) \quad (21)$$

This work practically applied the log implementation of the noise-aware α_{nehvi} acquisition function `qLogNoisyExpectedHypervolumeImprovement` (“qLNEHVI”) which has been shown to improve Bayesian optimisation performance in noisy settings⁷¹.

Implementation All Bayesian experiments were implemented in the Python3 programming language, with functionality spanning initial sampling techniques, adaptive sequential sampling methods, and an experimental workflow management system facilitating human-in-the-loop laboratory work; this package is given the acronym “BO4ACST” (version 1.0.0)- “Bayesian optimisation for atmospheric carbon sequestering thermosets”⁷². Note that Bayesian sequential optimisation techniques including Gaussian process regression, gaussian process conditioning, and sample utility evaluation were facilitated by the open-source adaptive experimentation platform or “Ax” (version 1.1.2)⁷³. Note that “Ax” makes

use of a great deal of code from “Botorch”, especially in its implementation of custom Bayesian optimisation routines (e.g. non-standard acquisition function implementations)⁷⁴.

3 Results

All results stem from the datasets generated during the practical-application of Bayesian optimisation in the domain of thermoset composite property development. Practical experimental results were split into those derived from the single-objective experiments and the singular multi-objective optimisation.

3.1 Single-Objective Experiments

Two single-objective Bayesian optimisation experiments were performed, each using a different objective variable for sequential sampling: Young’s modulus E or yield strength σ_y . Initial Gaussian process priors developed for both experiments were trained on the same eight quasirandomly sampled (Sobol) points $\mathbf{X}$ (from across the parameter space $\mathcal{X}$) and their evaluations $\mathbf{Y}$. Both experiments proceeded sequentially with a single sample evaluated at each Bayesian iteration. The sample limit was set to 24 (including the prior-forming dataset), and the optimisation was one of objective metric maximisation.

The objective-naive MIPT sampling technique was used to sample in a purely exploratory fashion, which provided a benchmark against which Bayesian optimisation routine progression could be gauged. Given its objective-naivety, purely exploratory nature, and the fact that it was initiated on the same prior-forming dataset as the Bayesian routines, the MIPT routine was only run once (samples having both Young’s modulus and yield strength evaluated). The resulting dataset could be directly com-

pared with both the single- and multi-objective Bayesian sequential routines (see Section 3.2).

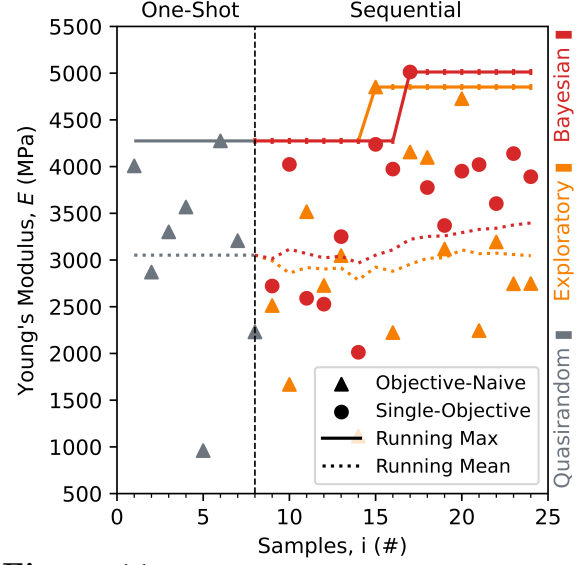


Figure 11: Progression of a the sequential optimisation experiment concerning the maximisation of Young's modulus. Initial one-shot samples drawn quasirandomly are presented separately from the two sequential techniques which they prime. These two sequential techniques are either objective-naive (MIPT) or single-objective (Bayesian).

Young's Modulus Experiment Figure 11 compares the single-objective optimisation progression of both Bayesian and exploratory sequential routines in the maximisation of Young's modulus. The one-shot quasirandomly-sampled initial dataset provided a baseline maximum of 4274 MPa and a mean of 3052 MPa. On reaching the sample limit, the single-objective Bayesian technique had exceeded both these baselines (max 5012 MPa, mean 3396 MPa) whilst the exploratory sampler only exceeded the maximum baseline (max 4851 MPa, mean 3046 MPa). Although the single-objective Bayesian routine ultimately discovered a sample exhibiting the highest Young's modulus within this experiment, it is interesting that the exploratory routine discovered a greater number of points surpassing the baseline maximum set by the initial dataset. The exploratory sampling routine was the first to locate a sample improving upon the baseline maximum: sample 15 with a Young's modulus of 4851 MPa. Subsequently, at sample 17, the Bayesian single-objective routine discovered the experiment's highest Young's modulus of 5012 MPa.

Yield Point Experiment Figure 12 compares optimisation progression of both the Bayesian single-objective and exploratory sequential routines with respect to the maximisation of the yield point. The one-shot quasirandomly-sampled initial dataset provided a baseline maximum yield point of 137 MPa and a mean of 123 MPa. On reaching the sample limit, the Bayesian technique had surpassed the baseline maximum (143 MPa) and remained level with respect to the mean (123 MPa), whilst the exploratory routine had surpassed the baseline maximum (140 MPa) and failed to improve upon the baseline mean (118 MPa). In this experiment both the Bayesian single-objective and exploratory routines discovered the same number of points surpassing the baseline maximum, although the Bayesian technique ultimately uncovered the sample with the highest yield strength. The exploratory sampling technique was the first to discover a point improving upon the maximum baseline; with this happening on sample 15, with a yield strength of 140 MPa. Subsequently, the Bayesian sequential tech-

nique improves upon the maximum baseline on sample 18 (140 MPa) and surpasses the exploratory routine on sample 21, with a final yield strength of 143 MPa.

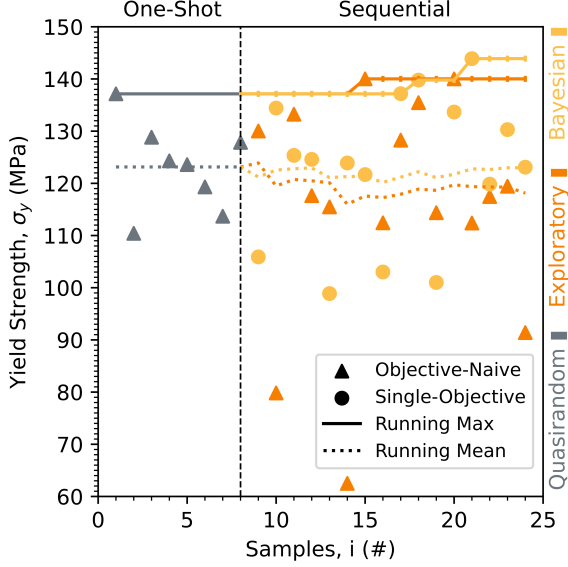


Figure 12: *Progression of a sequential optimisation experiment concerning the maximisation of yield strength. Initial one-shot samples drawn quasirandomly are presented separately from the two sequential techniques which they prime. These two sequential techniques are either objective-naive (MIPT) or single-objective (Bayesian).*

Comparison of Bayesian and MIPT Techniques In both single-objective optimisation experiments, the Bayesian sequential routines outperformed the exploratory sampling routine in terms of discovery of samples approaching an experimental global maximum within the allotted sample limit. Bayesian techniques benefit from adaptive sampling based on the balancing of both exploitative and exploratory behaviour, the former exhibited clearly in the non-decreasing trends shown by the Bayesian running means seen in both Figures 11&12. These results support the hypothesis that the Bayesian sampling routine was more effective in maximising the objective property during the course of a sequential optimisation, and although the margin of improvement was slim they remain significant. Further support may be garnered from repeat experiments of either a physical or computational nature.

However, in both experiments, the ex-

ploratory sampling routine outperformed the Bayesian single-objective technique in terms of sample efficiency: the exploratory technique improved upon the baseline maxima ($E = 4274$ MPa and $\sigma_y = 137$ MPa) of both experiments on drawing sample 15 for evaluation ($E = 4851$ and $\sigma_y = 140$). The fact that this occurred on the 15th sample for both experiments is trivially explained by the fact that this was the same exploratory routine being performed. As opposed to the two single-objective routines seen in Figure 11 and Figure 12 which were sequentially drawing unique sample in attempts to maximise different properties, the objective-naive MIPT sampler was only run once with each sequential sample characterised in terms of both target variables. This is why the especially well-performing sample 15 appears in both progression plots (Figs. 11/12). A hypothesis that the exploratory sampler more efficiently improves upon the baseline maximum than the Bayesian routine would be weakly supported by the above findings given that both circumstances were precipitated from the same data point, but further repeat studies in the physical or computational domain would aid in the supporting or negating this conjecture.

Styrenated Young’s Modulus Experiment An experiment identical to that of the single-objective Bayesian optimisation for Young’s modulus was carried out with the conventional reactive diluent styrene in place of the bio-based dimethyl itaconate. Figure 13 shows an optimisation progression plot for the styrene-based system. The single-objective Bayesian optimisation technique shows strong performance in terms of both sample efficiency and scale of improvement above the baseline. On conclusion of this sequential optimisation, the putative global optimum (sample 18) exhibited a compressive Young’s modulus of 7,271 MPa. Interestingly, the analogous bio-based experiment produced a sample (sample 17 in Fig. 11) exhibiting 69% the Young’s modulus

of this styrenated putative global optimum. Furthermore, when the relative crosslink densities between diitaconate and styrene based resins of 73% reported by Panic et al. are considered⁴², it becomes apparent that this short bout of optimisation has resulted in samples approaching the currently known bounds of mechanical performance.

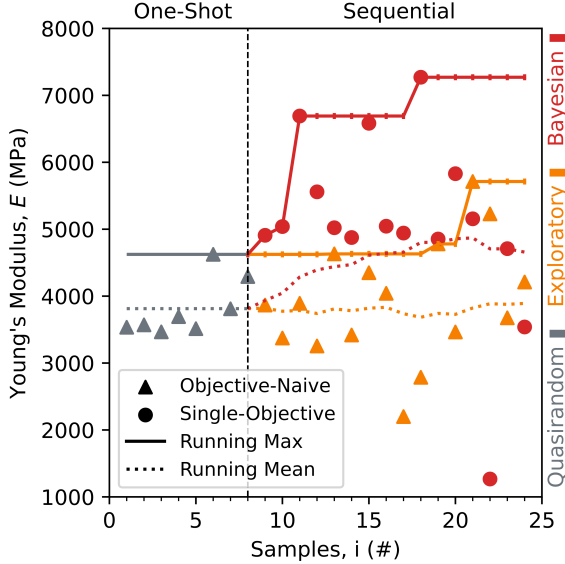


Figure 13: *Progression of a sequential optimisation experiment concerning the maximisation of Young's modulus with a styrene-based thermoset.*

3.2 Multi-Objective Experiment

In the multi-objective setting, a Bayesian sequential routine was performed which made use of the noisy expected hypervolume improvement acquisition function to sample-efficiently identify the Pareto front within an objective space comprised of both Young's modulus and yield strength. This dual-objective optimisation was facilitated by two separate Gaussian processes modelling the two different objective variables. Both were trained on 56 samples concatenated from the re-analysed raw data of the single-objective experiments described in Section 3.1. The order of concatenation (and re-indexing of sample numbers) was as follows: one-shot objective-naive quasirandom sampler (samples 1-8), sequential exploratory objective-naive routine (9-24), sequential Bayesian yield strength maximising

routine (25-40), and the sequential Bayesian Young's modulus maximising routine (41-56). This prior-forming dataset $\mathcal{D}$ yielded an initial Pareto front $\mathcal{P}$ comprised of three samples: two having been discovered as maxima by Bayesian sequential routines in their two respective single-objective experiments with yield strength (sample 37) and Young's modulus (sample 49) as their objectives. Whilst the third (sample 15) was derived from the exploratory technique as a point which performed well in both objectives of interest. With an ultimate sample limit of 72 and a single sample drawn per Bayesian iteration, the sequential multi-objective Bayesian routine proceeded to develop the Pareto front over 15 further samples (57-71).

All samples both prior-forming and sequentially-obtained are depicted in Figure 14, with the initial Pareto front marked as a dashed black line (Fig.14A). Given the initial Pareto front and considering only the direct evaluations obtained for samples it can be said that five samples were discovered which improved upon the baseline described by the Pareto front (samples 62,66,68,69,71). Under progression of the multi-objective routine, the first novel pareto optimal point which improved the Pareto front was the 62nd sample which dominated the 37th sample shifting the boundaries of the developing Pareto front (Fig.14B). Subsequently, the sample 66 evaluated ($E = 5043$ MPa and $\sigma_y = 153$ MPa) presented a marginal improvement upon both the Young's modulus of sample 49 ($E = 5012$ MPa) and the yield strength of sample 62 ($\sigma_y = 150$ MPa), collapsing the Pareto front to a single Pareto optimal point. Finally, the sample 68 dominated sample 66 producing the putative globally Pareto optimal solution ($E = 5458$ MPa and $\sigma_y = 165$ MPa), which may be contrasted starkly with the anti-Pareto front comprised of the 14th ($E = 1114$ MPa and $\sigma_y = 62$ MPa) and 25th ($E = 813$ MPa and $\sigma_y = 105$ MPa) samples. How the acquisition function likely navigated the parameter

space and came to the samples it discovered has been reviewed from the perspective of the final Gaussian processes developed and is appended in the supporting information.

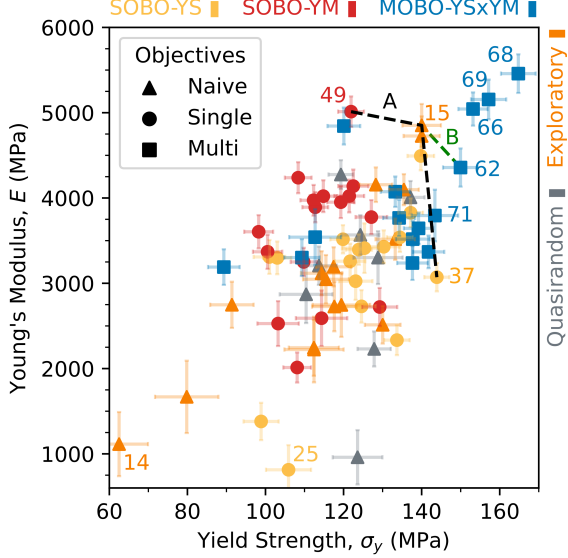


Figure 14: Results from a multi-Objective Bayesian optimisation routine. Bayesian confidence intervals were a single standard deviation and sampled from the final Gaussian processes. (A) Initial Pareto front as a dashed black line. (B) Pareto front improvement as a dashed green line. YS, Yield strength. YM, Young’s modulus. SOBO, Single-objective Bayesian optimisation. MOBO, Multi-objective Bayesian optimisation.

Figure 15 presents the strong relation between initiating agent content maximisation and improvements to sample optimality: the average content is 13% for the anti-Pareto front, 44% for the initial Pareto front, and 50% for the putative global optimum. Furthermore, the initiating agent’s composition also described a somewhat weaker relation between dibenzoyl peroxide marginalisation and sample optimality: anti-Pareto front 13.3%, initial Pareto front 12.3%, and 8% for the putative global optimum sample. No analogously clear trend existed with respect to the unsaturated polyester resin composition.

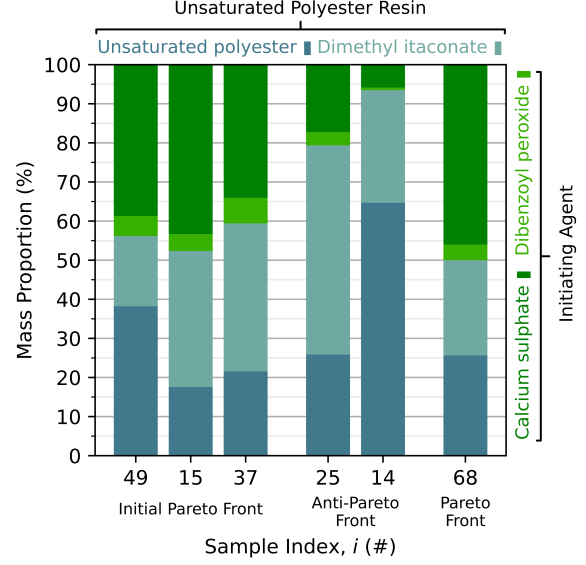


Figure 15: Compositional breakdown for samples of interest including the initial Pareto front, the final anti-Pareto front, and the final Pareto front.

4 Conclusion

This work shows that bio-based particulate-reinforced thermosets can be successfully optimised with respect to their compressive yield strength and Young’s modulus by adapting their composition during manufacture. Although filler content proved to be the means by which a large degree of strength was obtained, careful trade-off between unsaturated polyester and reactive diluent content was necessary in the high-filler region of the parameter space to obtain the globally optimal results. The detailed experiments show how the multi-objective Bayesian optimisation employing the log noisy expected hypervolume improvement acquisition function was capable of navigating this trade-off, producing a final Pareto front consisting of a single dominating point exhibiting $E = 5,458$ MPa and $\sigma_y = 165$ MPa.

In the domain of a sample-constrained single-objective Bayesian optimisation, using the log-noisy expected improvement acquisition function, a bio-based composite was obtained which exhibited 69% of the Young’s modulus of a conventional styrene-based material which had been optimised

in the same fashion. Cross-comparison of the aforementioned multi-objective putative global optimum sample with this styrene-based sample developed, showed that it attained 75% the conventional material’s strength. Although there exist no directly comparable studies concerning these differences, the extent to which this bio-based monomer has managed to substitute this conventional fossil-derived monomer is of remark, especially given its well-known tardiness in homopolymerisation³⁸.

The present study optimised compressive properties only; tensile, flexural, fracture, thermal, and ageing properties remain to be assessed. Replicate testing at selected formulations would further quantify experimental noise and improve $\mathcal{GP}$ noise estimation. Finally, although the resin chemistry is bio-derived, full sustainability assessment would require lifecycle analysis of monomer production, curing, filler sourcing, and end-of-life processing. Given the interesting functional relationships witnessed within the parameter space with respect to predictor x_3 , which concerns the balance between unsaturated polyester and reactive diluent, future work could examine how reactive diluents impact the mechanical properties of these materials. An optimisation of the styrene-based analogues in the single-objective setting with respect to the objective of yield strength would also be a worthwhile study.

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glass tube (Fig. 16E) and were captured in a distillation column (Fig. 16G) which was connected to a reboiler (Fig. 16F) at its base. The temperature at the top of the column (Fig. 16H) was maintained at around 70°C by adjusting the temperature of the reboiler (120°C) and the level of insulation applied around the column; this ensured a steady stream of methanol-rich vapours passing through the Claisen head (Fig. 16I) and re-condensing within the Liebig condenser (Fig. 16J). Methanol-rich liquid was there-with returned to the lower round-bottomed flask via a dropping funnel (Fig. 16K), whilst water was caught in the reboiler (Fig. 16F), pushing the reversible esterification reaction toward the desired diester product.

6 Supporting Information

Details of practical method and data analysis which cramped the presentation of material in the present paper have been gathered and elucidated within this section.

6.1 Notes on Synthesis of Di-itaconate

Dimethyl itaconate was synthesised by esterification of itaconic acid with methanol by the Fischer method, here we describe the practical method by which this was achieved. Figure 16 shows how this reaction mixture was placed in the lower round-bottomed flask (Fig. 16D), and brought to a vigorous boil at around 130°C. Vapours of both water and methanol passed up an insulated

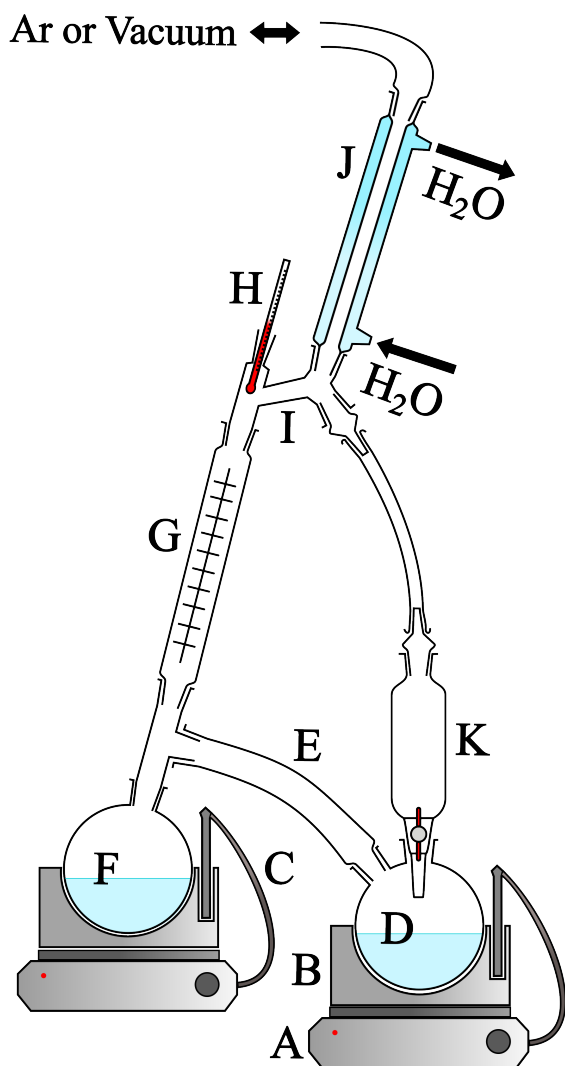


Figure 16: Apparatus for synthesis of dimethyl itaconate. (A) Hotplate stirrer (B) Aluminium bowl (C) Temperature controller probe (D) Three-necked round-bottomed flask containing the reaction mixture (E) Insulated glass tube (F) Single-necked round-bottomed flask acting as a distillation column reboiler containing water (G) Distillation column (H) Thermometer (I) Four-way Claisen head (J) Liebig condenser (K) Dropping funnel for recycled methanol

Before initiating the reaction by starting both the hotplate stirrers (Fig. 16A), an argon-rich atmosphere was generated by using a Schlenk line to cycle a vacuum (500 mbar) and flow of argon gas (350 mL min^{-1}); this was to discourage unwanted oxidation reactions and reduce the chances of vapour ignition. During the reaction however a valve was opened to allow pressure equilibration between the reaction vessels and the atmosphere in order to miti-

gate the risk of explosion. Methanol cycling was halted 3.5 h into the reaction by closing the dropping funnel (Fig. 16K), after which all heating was ceased on reaching 4 h.

Having trialled several solvents (e.g. cyclohexane, n-hexane), toluene was found to be the most suitable as its relatively higher affinity for dimethyl itaconate during liquid-liquid extraction led to acceptable yields at conclusion of workup. The final product mixture retrieved from the round-bottomed flask (Fig. 16D) was placed in a separating funnel with toluene in a 1:3 volume ratio. The separating funnel contents were washed with water three times, saturated sodium bicarbonate solution twice, and saturated sodium chloride once; each washing step involved a volume ratio of 2:1 between the previously obtained organic fraction and the next volume of washing liquid used. A gravity filtration step was then carried out between the water and sodium bicarbonate washing steps to remove insolubles. Following washing, the organic fraction was removed to an Erlenmeyer flask and dried using anhydrous magnesium sulphate; thereafter removed using gravity filtration. Dry organics were placed in a rotary evaporator and heated at 40°C whilst the pressure was reduced from an initial 70 mbar to 20 mbar over an average period of 8 h; this removed all but a small mass of toluene and water from the liquid product. From this product, dimethyl itaconate was crystallised out at room temperature by addition of a nucleating agent. The nucleating agent was crystalline dimethyl itaconate obtained directly from a particularly pure batch of rotary evaporation product which had crystallised in a fridge at 3°C . The above method afforded a 45% yield of product, although higher values may be possible if an extended period of boiling were adopted.

6.2 Notes on the TLC of Diitaconates

Initial estimates of purity were garnered from spotting chemicals onto thin layer chromatography plates (polyester with a layer of silica gel and ultraviolet indicator; 6 mm pore diameter and 0.2 mm layer thickness) and using a mixture of dichloromethane and methanol (2.5:7.5 v/v respectively) as the mobile phase. On completion, spot(s) could be observed under ultraviolet light ($\lambda = 254$ nm) and their retardation factors (R_F) calculated using equation 22, relating this metric with the distance migrated by the substance in question (b in mm) and that travelled by the mobile phase (a in mm)⁷⁵.

$$R_F = \frac{b}{a} \quad (22)$$

Having solvated and diluted pure itaconic acid to a 1.25% w/w solution using stock mobile phase, this solution was spotted on a plate- this produced a single spot with an R_F value of 0.79. Meanwhile, having spotted a 5% w/w solution of the final crystalline product (expected to be dimethyl itaconate), a single spot with an R_F of 0.91 was obtained. These results provided benchmarks by which reaction progress could be gauged as well as supporting the hypothesis that these two crystalline materials were reasonably pure, although the presence of non-fluorescent or trace chemicals cannot be ruled out using TLC alone.

Samples taken periodically from the reaction vessel during the course of reflux revealed a change in composition: the first sample (retrieved after 15 min of heating) presented a single spot with an R_F value of 0.81, while a final sample (retrieved after 4 h) presented two spots with R_F values of 0.90 and 0.81 respectively. Given the similarity of these samples with those of the crystalline reactant and product analysed previously, these results suggest that partial conversion of the reactant itaconic acid into the desired product dimethyl itaconate was achieved.

6.3 Notes on Synthesis of Unsaturated Polyester Resin

Figure 17 shows the apparatus used to synthesise the unsaturated polyester. A Dean-Stark trap (Fig. 17I) was used to allow the ‘azeotropic method’ of polyester synthesis (as opposed to the ‘dry’ method) which reduced the time taken to synthesise a high molecular weight polyester⁵⁴. In this method, an entraining agent (mixed xylenes) which can form an azeotrope with the water was added to the mixture (5% w/w)^{56,76}. This increases the rate at which water is condensed and captured by the Dean-Stark trap, driving the reaction forward by removing the unwanted byproduct of water from the reaction mixture. Furthermore, argon gas was used to sparge the reaction mixture (Fig. 17E) facilitating the removal of water from the bulk mixture and mitigating the combustion risk associated with high temperature xylenes⁵⁴.

The synthesis procedure was initiated by charging the four-necked round-bottomed flask (Fig. 17D) with the requisite masses of the two monomers and xylenes, opening the argon taps to provide a flow of sparging gas at the rate of 5 bubbles per second through the argon injector (Fig. 17E)⁵⁴, opening the water taps to actively cool the Liebig condenser (Fig. 17I), setting the overhead stirrer to a speed of 50 rpm (Fig. 17G)⁵⁴, and bringing the reaction mixtures temperature (Fig. 17F) up to 160°C using the hotplate (Fig. 17A). During the procedure the mixture’s temperature was raised monotonically by 10°C h⁻¹ until it reached a maximum of 210°C, at which it was held.

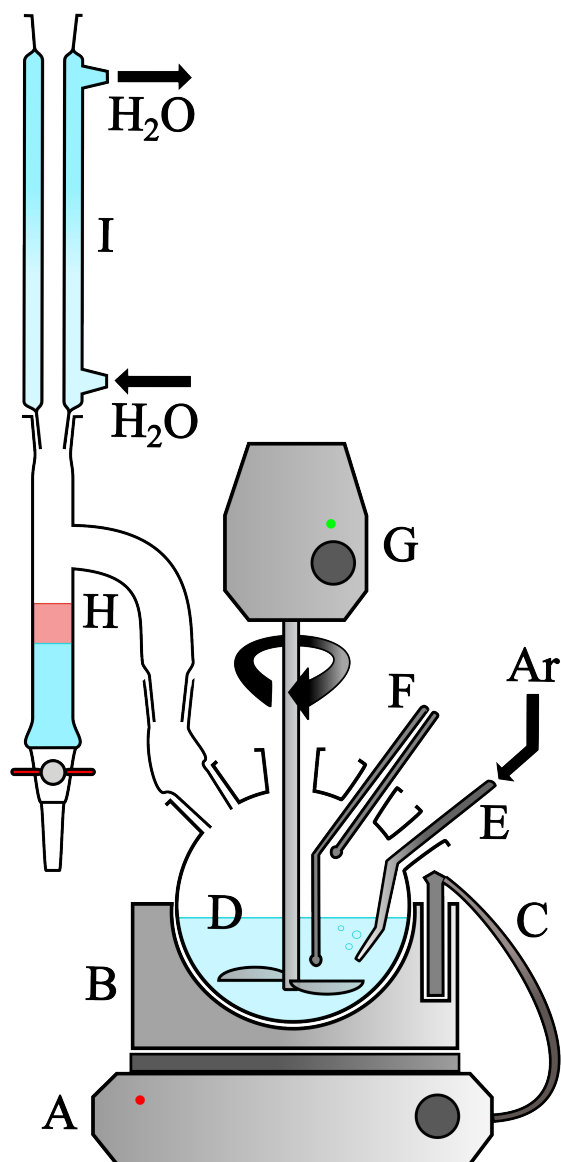


Figure 17: Apparatus for synthesis of unsaturated poly(propyl itaconate). (A) Hotplate stirrer (B) Aluminium bowl (C) Temperature controller probe (D) Four-necked round-bottomed flask containing the reaction mixture (E) Stainless-steel argon injector (F) Probes for batch and vapour temperatures (G) Overhead stirrer (H) Dean-Stark trap (I) Liebig condenser.

The reaction progress was ascertained by periodically retrieving a sample of the reaction mixture and ascertaining its acid number, in units of mg g^{-1} , describing the amount of potassium hydroxide required to neutralise one gram of sampled unsaturated polyester. This approximate form of end-group analysis is commonly used in unsaturated polyester synthesis^{40,41,43,54} and provides a proxy value estimating the number of un-esterified acid groups remaining within

the unsaturated polyester. An American Society for Testing and Materials standard (“ASTM D4662-20”) was used for this purpose. In-line with common practice, heating of the polyesterifying mixture was ceased at an acid value of 50^{54,56,76}. On cessation of heating, an outlet was opened for egress of remnant xylenes and the unsaturated polyester was allowed to cool in an air bath.

Since the viscosity of a cooling unsaturated polyester will increase monotonically until it reaches room temperature, where it exists as a solid thermoplastic mass, it is common to prepare the unsaturated polyester resin immediately after cessation of heating by adding reactive diluent directly to the cooling unsaturated polyester⁵⁴. Unfortunately, reactive diluents are liable to polymerise during this addition causing a premature gelation. Mitigation against hardening into thermoplastic or premature gelation into rubber can be achieved by addition of an addition-polymerisation inhibiting agent at a suitable intermediate temperature^{24,54,55}. In this work, we awaited a temperature of 100°C before adding the below specified amounts of inhibitor and reactive diluent.

Here, we used the commonly-used free-radical scavenger hydroquinone as an inhibiting agent, which was added so as to make up 0.06% w/w of the cooling unsaturated polyester mixture. Hydroquinone was dissolved in a small amount of propylene glycol before addition to ensure homogenous mixing into the bulk⁵⁴. The inhibitor percentage was determined through a process of trial-and-error, where lower percentages tended to facilitate premature gelation of unsaturated polyester resins on the shelf, whilst higher percentages entirely prevented the intended curing procedures for these resins. The percentage of inhibiting agent was above that which is commonly used in the synthesis of conventional unsaturated polyester resins (approximately 0.02%)^{54,55}. This was a consequence of both the relatively higher

proportion of unsaturated sites along the polyester backbone and a higher copolymerisation affinity between the reactive diluent and the sites of unsaturation.

Following introduction and homogenisation of inhibiting agent into the unsaturated polyester mixture, the reactive diluent (dimethyl itaconate) was added such that it made up 30% w/w of the final unsaturated polyester resin.

6.4 Notes on Gel Permeation Chromatography

In a separate experimental series (not considered in the context of the present work), a similar polyester was synthesised with a reduced content of itaconate moieties in its backbone. This was achieved by replacing some of the itaconic acid used during synthesis with succinic acid, producing poly(propylene glycol succinate itaconate). Figure 18 depicts the GPC results obtained on analysing this alternative formulation. On comparison with the GPC results obtained for poly(propylene glycol itaconate) as detailed in the present work it becomes apparent that this alternative formulation exhibits little if any premature crosslinking and associated network polymers. As the closest approximation to the distribution associated a poly(propylene glycol itaconate) polyester that didn't suffer network polymer genesis, we re-scaled the tail section of Figure 18 and inserted this into the apparent distribution for poly(propylene glycol itaconate) presented in the present work. This allowed the calculation of both number and weight average molecular weights for the hypothetically uncrosslinked polyester.

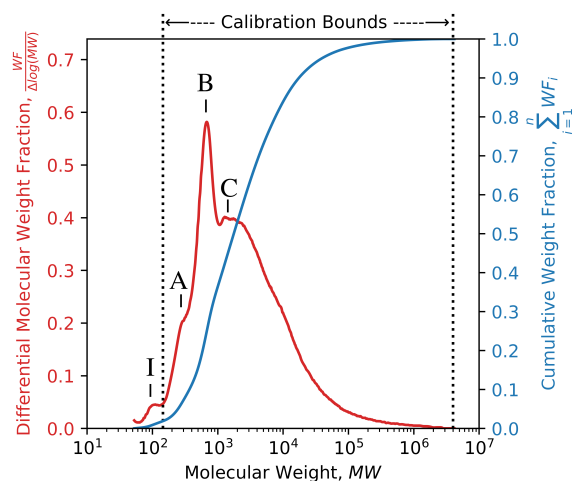


Figure 18: A graph depicting the distribution of mass across various polymer chain lengths for an alternative polyester sample tested using GPC.

An alternative line of investigation undertaken with respect to the GPC of these polyesters was that associated with the deconvolution of the apparent distribution. By identification of three physical peaks (Figure 4A/B/C) mathematical deconvolution was undertaken by fitting a log normal distribution to the three curves in isolation (by cropping the apparent distribution suitably), before fitting the three in unison (in an additive fashion) to the apparent distribution. Figure 19 shows the three deconvoluted peaks and their fit relative to the apparent distribution.

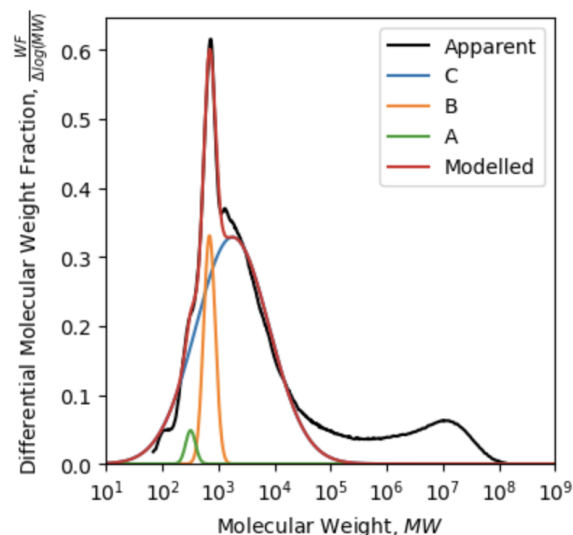


Figure 19: A graph of all three deconvoluted peaks which were identified from within the apparent distribution.

Having deconvoluted the three peaks and mused on the possibility of cyclisation as

the mechanism behind the existence of the two smaller peaks, we investigated various cyclisation mechanisms. Common to all polyesters are the ring closure and back biting mechanisms⁷⁷ whilst Ordelt reactions have been identified as particularly possible in the case of unsaturated polyesters incorporating itaconate moieties⁶². Since back biting and ring closure both result in the same structure we only modelled the latter. Figure 20 visualises ring closure: Fig 20A shows a linear section of polymer with a start length L_0 of 1, Fig 20B shows a perfectly circular loop with a maximum length L_1 of 0.32, whilst Fig 20C considers the extreme scenario in which the loop flattens entirely producing a maximum length L_1 of 0.5.

Ring Closure Cyclisation (RCC)

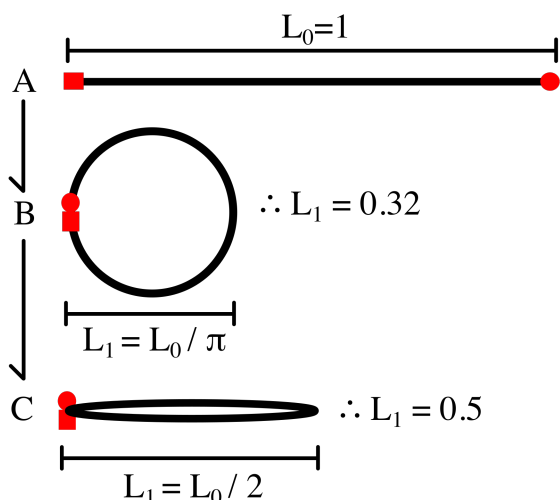


Figure 20: A schematic describing the model employed when considering the ring-closure cyclisation mechanism.

Figure 21 visualises Ordelt-based cyclisation: Fig 21A shows the linear section of polymer with an initial length of $L_0 = 1$, whilst an annotation below indicates the random nature by which an end hydroxyl group might react with an itaconate unsaturation anywhere along the polymer’s backbone. Fig 21B describes a perfectly circular loop situation where an end group has reacted at position $L_1 = 0.75$, producing a final maximum length L_2 of 0.92. Fig 21C describes the extreme scenario for the previous hypothetical $L_1 = 0.75$ situation, where the loop flattens completely and generates a final

maximum length $L_2 = 0.88$. Fig 21C1/C2 provide the two extreme scenarios of ring flattening at both ends of the possible range of L_1 values, providing the ultimately highest and lowest final maximum lengths possible under the Ordelt reaction mechanism $L_2 = [0.99, 0.51]$.

Ordelt Cyclisation (OC)

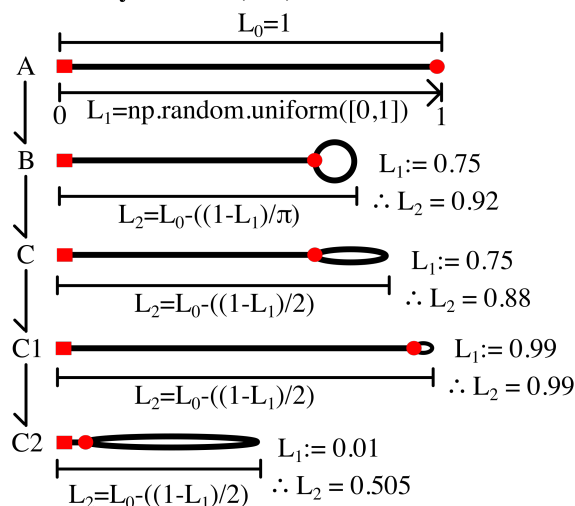


Figure 21: A schematic describing the model employed when considering the Ordelt cyclisation mechanism.

Given the set initial length $L_0 = 1$, Figure 22 describes the estimation of hydrodynamic volume for each of the scenarios (linear, Ordelt cyclised, and ring closure cyclised). Here, for each of the modelled scenarios we assumed conformations with the highest maximum lengths, we also assumed an average length between the maximum and minimum situations for the Ordelt scenario, whilst proxied hydrodynamic volumes were calculated by assuming the swept spherical area of conformations. A simple comparison of the final hydrodynamic volume proxy V_h values shows a clear hierarchy between the linear ($V_h = 1.05$), the Ordelt ($V_h = 0.59$), and the ring closure ($V_h = 0.26$) scenarios. Since the volume scales as a power of the dimensions, small changes in the total final length of the conformation could possibly cause drastic changes in the hydrodynamic volume. Figure 22 shows how a polyester molecule of the same molecular weight (modelled here as length) can exhibit a variety of hydrodynamic volumes due to

both cyclisation mechanisms and coformations adopted thereafter.

6.5 Notes on Mechanical Test Data Processing

Here is an unabridged overview of all the data processing undertaken during the mechanical testing of samples.

Hydrodynamic Volume Estimation

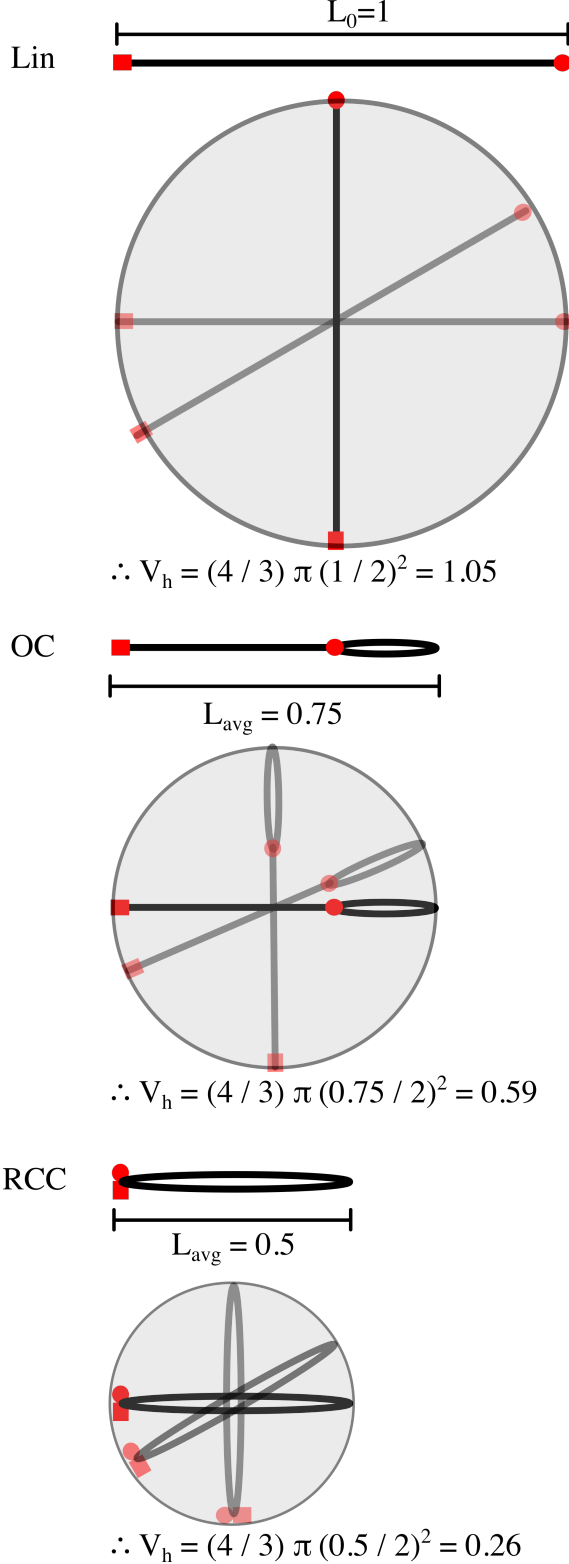


Figure 22: A schematic describing the simple model by which hydrodynamic volume of polymers was estimated.

Euler Buckling To mitigate against concerns of Euler buckling taking place in samples of the proposed dimensions, we calculated the critical load based on Euler's critical load equation and some estimated properties for conventional polyester thermosets⁶³. Equation 23 shows how the critical load (P_{cr} in N) is related to the elastic modulus of the material (E in Pa), the second moment of area (I in m^4), the effective length factor (K , unitless), and the initial length of the column in question (L in m).

$$P_{cr} = \frac{\pi EI}{(KL)^2} \quad (23)$$

Young's moduli for conventional thermoset polyesters were expected to range from 2-4.5 GPa²³. The second moment of the area for the top surface of the cylindrical sample with radius 6.35 mm was calculated to be $1.276 \times 10^{-9} m^4$ by assuming it to be a perfect filled circular area, for which the generic equation 24 applied (where the r is the radius of the cylinder in m).

$$I_y = \frac{\pi r^4}{4} \quad (24)$$

By assuming that both ends of the column would be braced against rotation or translation (given their compression between two platens), the theoretical column effective length factor K would have been 0.5; as a further precaution a conservative design value of 0.65 was assumed based on pertinent literature⁷⁸. On these assumptions it was estimated that the samples in question would be likely to undergo Euler buckling at an applied load of between 433-976 kN. As the compressive strength expected of conventional unsaturated polyester thermosets

are expected to fall within the range of 89-252 MPa, corresponding to maximum compressive forces of between 11-32 kN (given the surface area of the samples), we concluded that the samples were unlikely to experience Euler buckling.

Correction of Raw Data Finite compliance of the testing machine introduced an additional displacement contribution into raw exported datasets. To correct for this, a blank compression test was performed with no sample positioned between the platens, generating a force-displacement curve corresponding to the inherent compliance of the machine. This curve was subsequently used as a correctional dataset. Equation 25 shows how the corrected displacement value (d in mm) was obtained by taking a raw displacement value (d_{raw} in mm) and subtracting the corresponding correctional displacement value (d_{cor}).

$$d = d_{raw} - d_{cor} \quad (25)$$

Processing of Corrected Data Load values from a sample's corrected dataset were converted into engineering stress values by dividing by the initial surface area. Each cylindrical sample's diameter was measured using a "Magnusson" digital caliper (accuracy ± 0.02 mm) from three different positions (ϕ_1, ϕ_2, ϕ_3 in mm), each offset by roughly $+45^\circ$ around the circumference, and the mean of these values (ϕ_{avg} in mm) calculated (see Equation 26).

$$\phi_{avg} = \frac{\phi_1 + \phi_2 + \phi_3}{3} \quad (26)$$

Equation 27 shows how the initial top-face surface area was calculated from the average top surface diameter ϕ_{avg} .

$$A_{init} = 0.25 \times \pi \times \phi_{avg}^2 \quad (27)$$

Subsequently, the engineering stress (σ in MPa) could be calculated (Equation 28) by

considering the force applied (F in N) and the initial top surface area, A_{init} ⁶⁴.

$$\sigma_{eng} = \frac{F}{A_{init}} \quad (28)$$

Displacement values from a sample's corrected dataset were converted into engineering strain by taking the difference between the initial length of the samples and the length of the sample during compression. The initial length (l_{init} in mm) of sample was calculated by taking the mean of three length measurements (l_1, l_2, l_3 in mm) (these taken in the same fashion as for diameter) as shown in Equation 29.

$$l_{init} = \frac{l_1 + l_2 + l_3}{3} \quad (29)$$

Equation 30 shows the simple link between length of a sample at a given point during a compressive test (l in mm), the initial sample length (l_{init}) and the corrected displacement at that point d .

$$l = l_{init} - d \quad (30)$$

Subsequently, the engineering strain was calculated by finding the absolute change in sample length (Δl_{init} in mm) and dividing this by the initial length of the sample (l_{init}) as shown in Equation 31.

$$\epsilon_{eng} = \frac{\Delta l_{init}}{l_{init}} = \frac{l - l_{init}}{l_{init}} \quad (31)$$

Stress-Strain Smoothing To produce a smoothed dataset (from which Young's modulus and yield strength could be deduced) a combination of sample-rate reduction and filtering were applied to downsample the stress-strain dataset. Downsampling was achieved using a windowed averaging technique, whereby the unsmoothed dataset (p_{us}) was split into a number of windows along the strain-axis (p_s) by electing a suitable number of points per window (p_w), as shown in Equation 32. For convenience, p_{us} is required to be divisible by p_w , and $p_s > 200$ was found to give reliable results. Given

the constant size of the unsmoothed dataset ($p_{\text{us}} = 1,000$), a constant number of points per window was elected ($p_w = 5$).

$$p_s = p_{\text{us}}/p_w \quad (32)$$

Equation 33 shows how the process of averaging stress (σ_{eng}^i) and strain (ϵ_{eng}^i) values within a given window yielded a single point described by its average stress ($\bar{\sigma}_{\text{eng}}^j$) and strain ($\bar{\epsilon}_{\text{eng}}^j$).

$$\bar{\sigma}_{\text{eng}}^j, \bar{\epsilon}_{\text{eng}}^j = \frac{\sum_{i=1}^{p_w} \sigma_{\text{eng}}^i}{p_w}, \frac{\sum_{i=1}^{p_w} \epsilon_{\text{eng}}^i}{p_w} \quad (33)$$

Averaged points from each window were assembled to form the new downsampled dataset depicted in Equation 34.

$$[[\bar{\sigma}_{\text{eng}}^j, \bar{\epsilon}_{\text{eng}}^j], [\bar{\sigma}_{\text{eng}}^{j+1}, \bar{\epsilon}_{\text{eng}}^{j+1}], \dots, [\bar{\sigma}_{\text{eng}}^{p_s}, \bar{\epsilon}_{\text{eng}}^{p_s}]] \quad (34)$$

Figure 23 demonstrates the efficacy of this technique in removing noise from the unsmoothed stress-strain dataset.

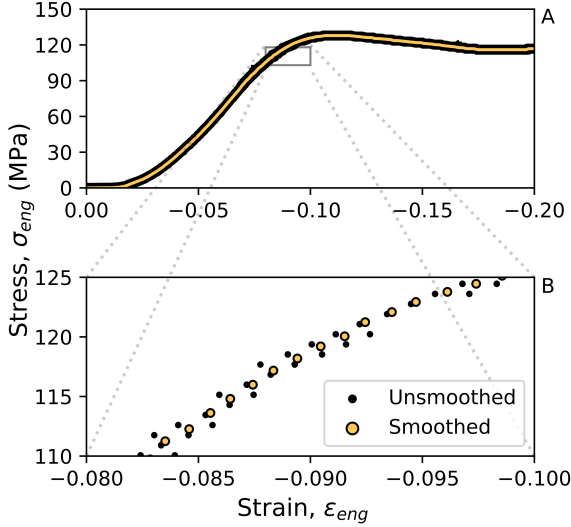


Figure 23: Multi-scale depiction of the stress-strain dataset before and after downscaling. (A) Overview line plot. (B) Zoomed scatter plot.

Derivative Analysis The proportional limit is defined as the stress at which a material ceases to obey Hooke's law (i.e. stress no longer increases proportionally with strain). The gradient of the preceding elastic region of deformation is generally defined as the Young's modulus. However, given that the stress-strain curves did not consistently exhibit a linear region, the limit of proportionality was equated to the point at which

the stress-strain curve began to level off (i.e. the first point of inflection). To find this inflection point, the first derivative of stress with respect to strain (i.e. tangent modulus E_t in MPa) was calculated from the smoothed stress-strain dataset by taking into account the instantaneous change in both stress ($\Delta\sigma_{\text{eng}}$) and strain ($\Delta\epsilon_{\text{eng}}$), and given by Equation 35.

$$E_t = \frac{\Delta\sigma_{\text{eng}}}{\Delta\epsilon_{\text{eng}}} = \frac{\sigma_{\text{eng}}^{j+1} - \sigma_{\text{eng}}^j}{\epsilon_{\text{eng}}^{j+1} - \epsilon_{\text{eng}}^j} \quad (35)$$

Figure 24 shows how smoothing of the stress-strain dataset facilitated the analysis of the resultant tangent modulus versus strain dataset. Step-like features are apparent in the tangent-modulus curve obtained from the unsmoothed stress-strain data. These may be consistent with either distretisation or internal processing of the exported displacement signals and subsequently amplified by the process of numerical differentiation. To avoid undue physical significance being assigned to these artefacts, proportional-limit detection was performed upon a smoothed stress-strain dataset.

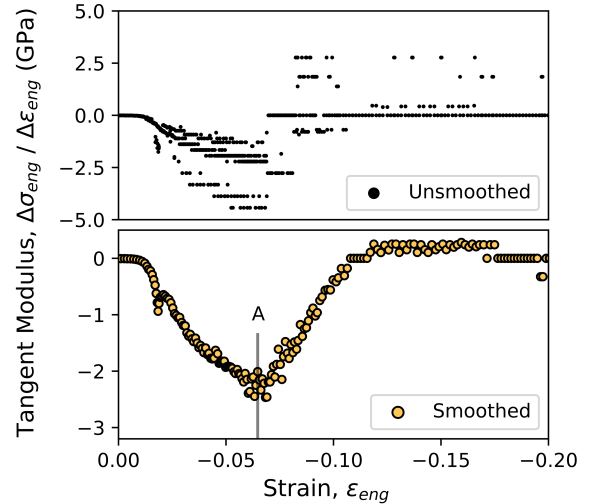


Figure 24: A comparison of the tangent modulus vs strain curves obtained from both the raw and the smoothed counterpart stress-strain datasets. (A) Strain at first inflection point.

Detection of the first inflection point was automated using an algorithm that iterated chronologically through the tangent modulus versus strain dataset from low to high strains. At each point, the algorithm cal-

culates an average gradient exhibited by an equal number of points both before and after the current strain value (the ‘point horizon’). Using a point horizon of ten, the first inflection point was defined as the first point at which the summed forward and backward gradients fell below zero. This point was taken as the proportional limit, as shown in Figure 24A.

6.6 Notes on Gaussian Processes

Figure 25 presents a matrix of slice plots involving the two Gaussian processes as they existed at the culmination of the multi-objective experiment. Every plot holds $2/3$ predictors constant whilst the column dictates which of the three predictors is varied. Each row of the matrix conditions the Gaussian process $\mathcal{GP}$ on a different set of predictor constants, these are derived from either the 14th sample ($\mathbf{x}^{i=14}$), the 68th sample ($\mathbf{x}^{i=68}$), or the mean of all samples’ predictors ($\mathbf{x}^\mu$). The graphs are further annotated with vertical lines representing sample predictor values ($\{x_1, x_2, x_3\}$): $\mathbf{x}^{i=14} = \{0.98, 0.5, 0.93\}$ $\mathbf{x}^{i=68} = \{0.72, 0.4, 0.5\}$ $\mathbf{x}^\mu = \{0.71, 0.64, 0.64\}$.

Row B of Figure 25 describes the processes as conditioned upon the experimental mean of sample predictors ($\mathcal{GP}|\mathbf{x}^\mu$); these graphs provide rough intuition with respect to the objective topography of the parameter space but naturally fail to account for why specific combinations of predictor values (e.g. sample 68 or 14) exhibit higher or lower performance in both the metrics of interest.

In Figure 25A we see the processes conditioned on the 68th sample’s predictors $\mathcal{GP}|\mathbf{x}^{i=68}$: in the case of predictor x_3 ’s yield strength process (Fig.25:A3) we see a topographical deviation from the mean conditioned process (Fig.25:B3). This deviation was a flattening in the region where $x_3 = [0.5, 0.65]$ which shifted the local optimum toward $x_3 = 0.5$: this resulted in

an alignment of both local optima of the two processes, which partly explains why marginalisation of UPR content by sample 68 leads to improved performance in both objective metrics.

Although the troughs apparent in the mean processes with respect to x_1 and x_3 (Fig.25:B1&B2) hinted at why the 14th sample’s predictor value combination produced suboptimal performance in both the objective metrics, the x_2 topography failed to do so. On examination of the x_2 topography of the processes conditioned on the 14th sample’s predictors $\mathcal{GP}|\mathbf{x}^{i=14}$ we find large deviations from the mean processes: both the topographies have fallen noticeably, the Young’s modulus process has flattened, and the yield strength process inverted such that the local optimum shifted from $x_3 = 0.4$ to $x_3 = 1$. Furthermore, this inversion presented a local trough in the vicinity of the 14th sample explaining further why this sample behaved suboptimally.

These three observations are respectively reflected in the Gaussian process topographies presented in Figure ??: when considering predictor x_3 (column 3: controlling balance of unsaturated polyester resin to initiating agent) we see that Gaussian processes clearly describe a negative correlation between x_3 and optimality in all three slice plots (A3,B3,C3). These strong correlations and coinciding local optima between $\mathcal{GP}$ ’s would explain why there is a clear global link between the balance of unsaturated polyester resin and initiating agent and sample optimality, With respect to x_2 (column 2: controlling initiating agent composition) the $\mathcal{GP}$ ’s display a negative correlation between x_2 and optimality in two of the slice plots (A2,B2), but this relation disappears and is replaced with a flattened and lowered process in at least one region of the parameter space (C2). In the case of x_1 (column 1: controlling unsaturated polyester resin composition) we can see that the $\mathcal{GP}$ ’s display a function shape with singular peaks at varying positions in all the different slice

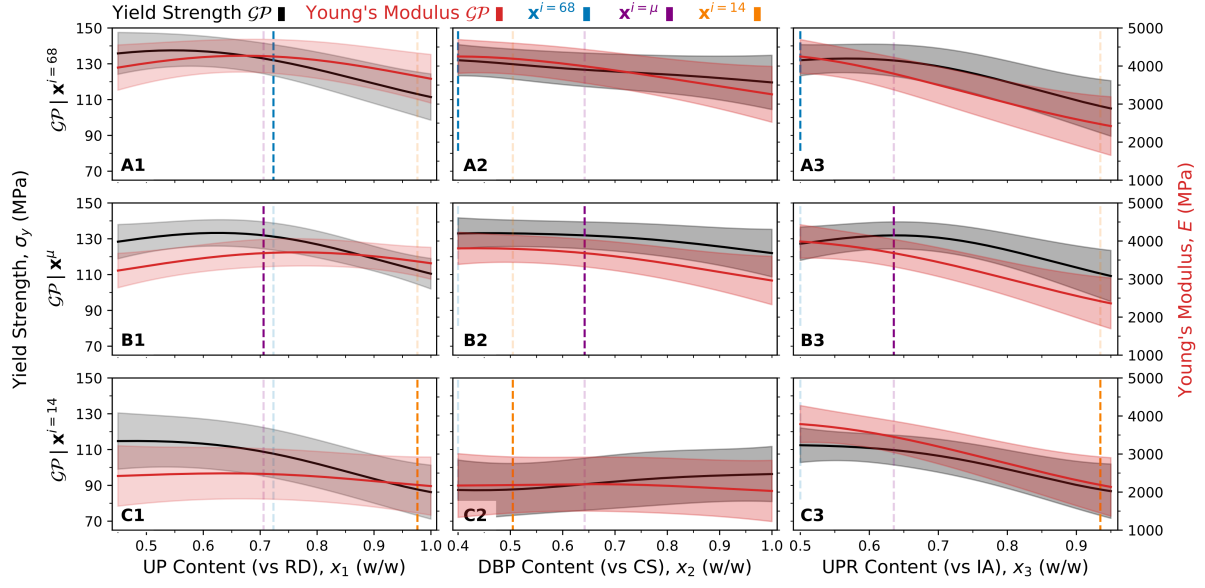


Figure 25: Slice plot of the Gaussian processes developed by the multi-objective experiment with mean function and a 95% confidence interval (1.96σ) presented. UP, Unsaturated polyester. RD, Reactive diluent. DBP, Dibenzoyl peroxide. CS, Calcium sulphate. UPR, Unsaturated polyester resin. IA, Initiating agent.

plots (A1,B1,C1); this added optimisation polyester resin content with sample optimal-complexity explains well why there is not ob- ity.
vious global trend relating the unsaturated